ASTR 302 Fall 2026
Characteristics of Spectrographs



IGRINS Spectrograph (Park+ 2014)
SPIE Paper

Agenda – Oct 5, 2026

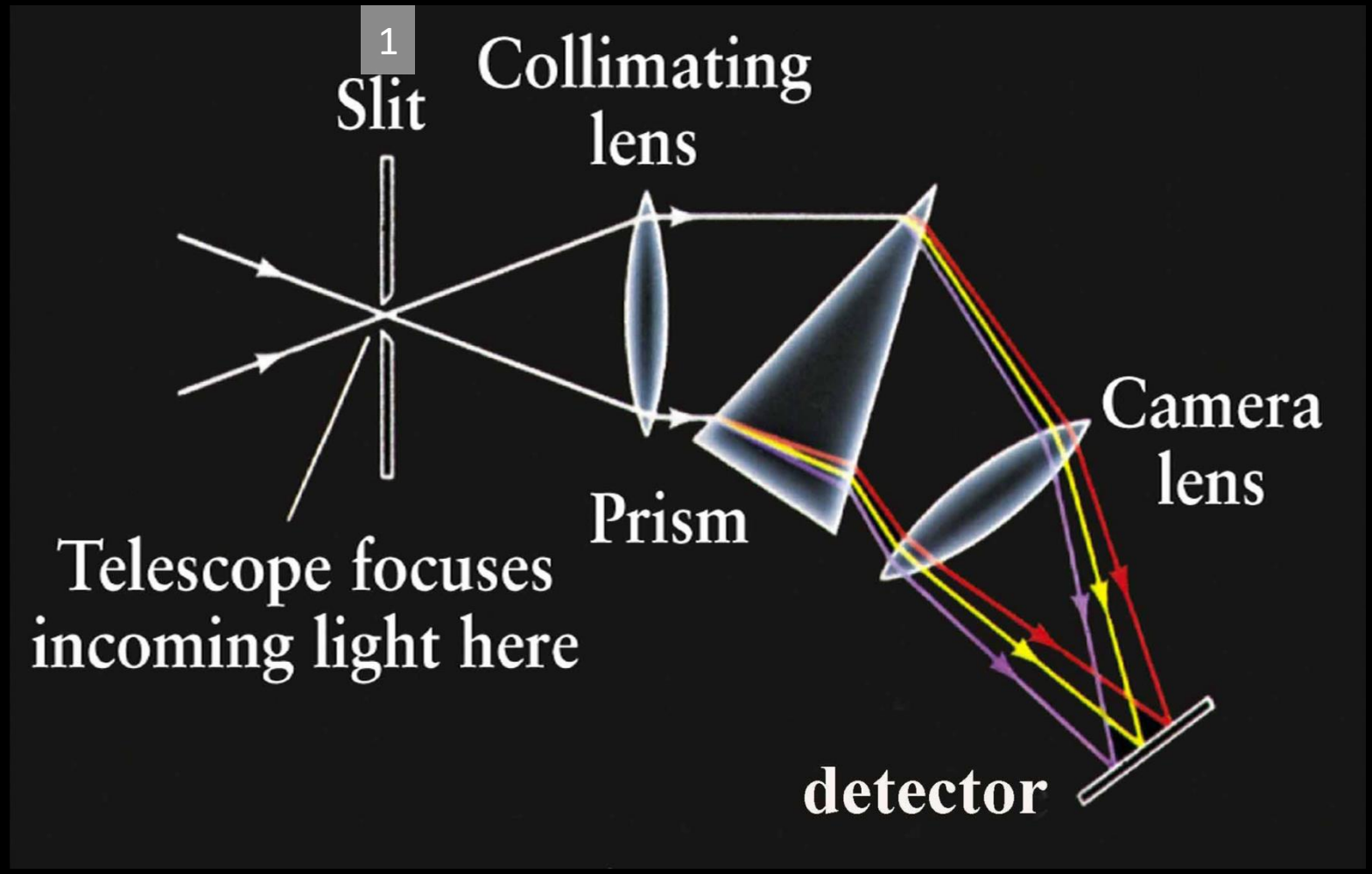
- Weekend Recap
- Photometry Lab due Friday October 9th
- Spectrographs (Dispersers)

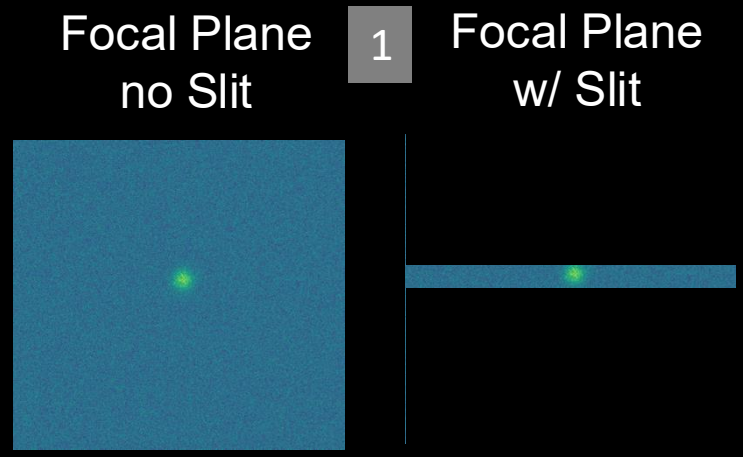
Recommended Reading: Birney Chapter 12

Weekend RECAP!

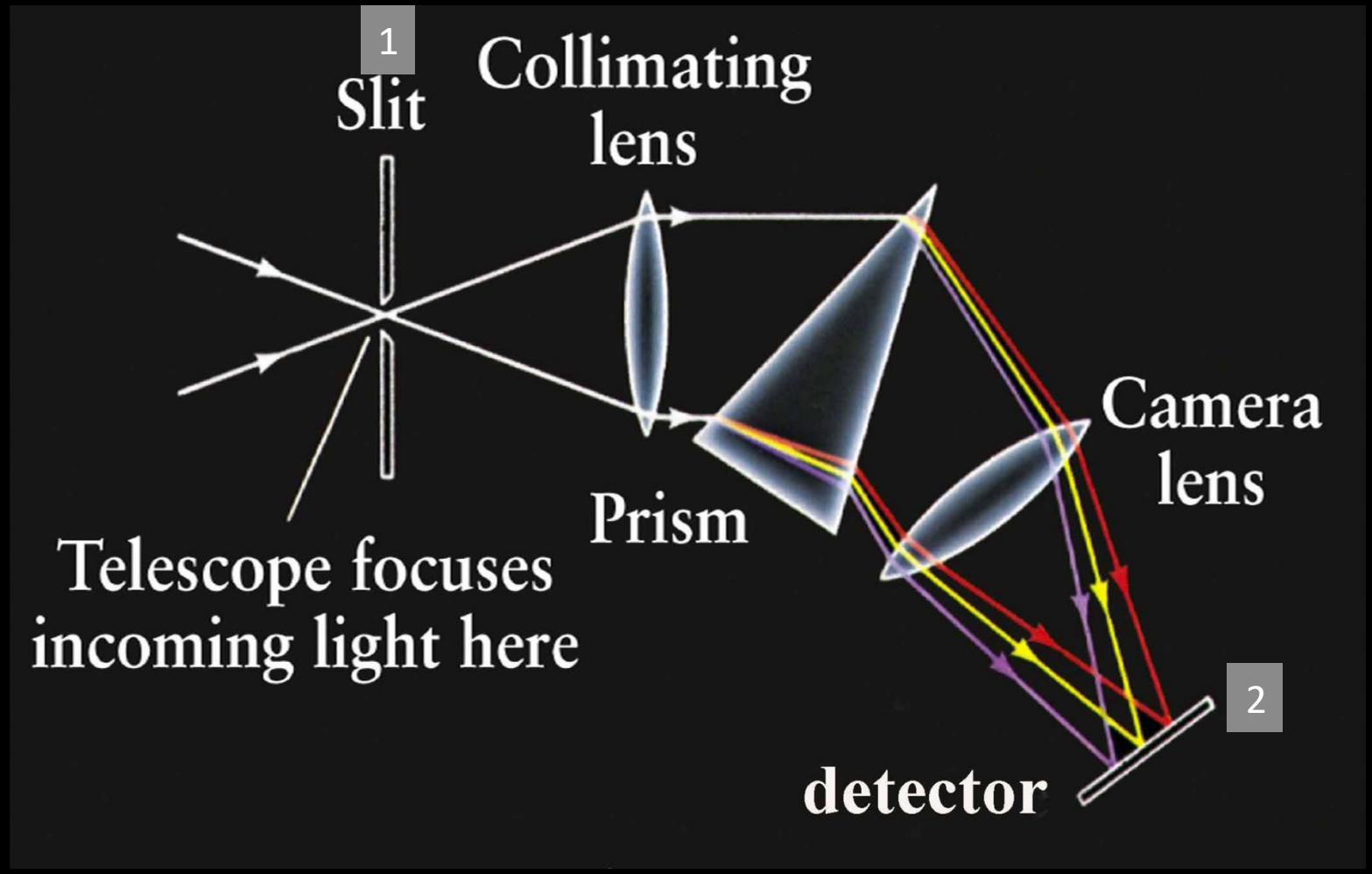


Components of a Spectrograph

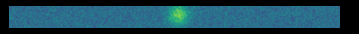




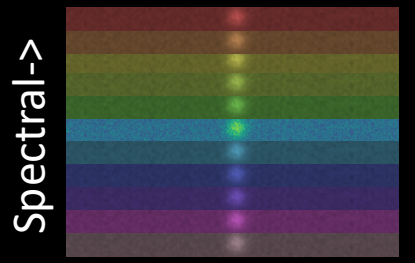
Components of a Spectrograph



Focal Plane w/ Slit 1



Detector 2



Characteristics of Spectrographs

Dispersion Power – How much different wavelengths are spread out. This is determined by whatever dispersive element(s) is used.

Resolution – How well your instrument can measure wavelength differences. This is determined by all your optics and detector together. This is usually what an observatory or instrument manual would report.

Types of Dispersive Elements

- **Prism** (Transmissive)
- **Diffraction Grating** (Transmissive or Reflective)
- **Grism** (Prism + Grating)

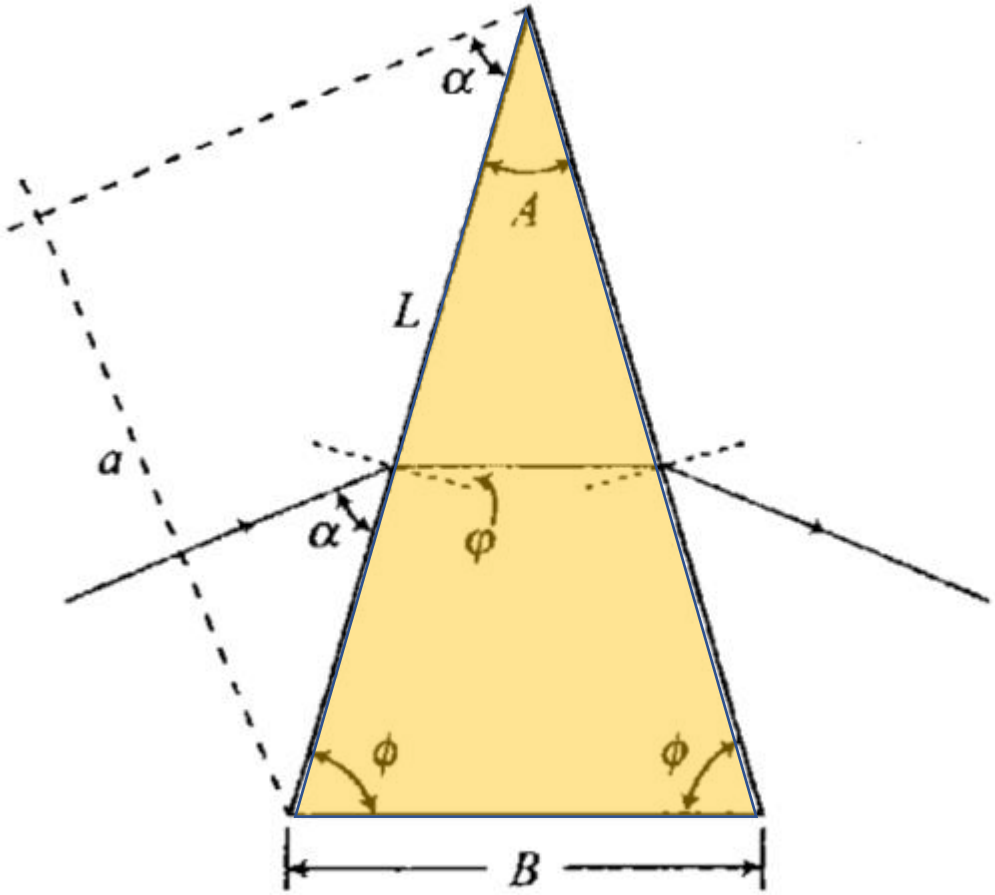
Types of Dispersive Elements: Prisms

- **Prisms** disperse light by capitalizing on the fact that refraction is a wavelength dependent phenomena.

- The dispersion power of a prism depends on the shape of the prism and the material's index of refraction n(λ):

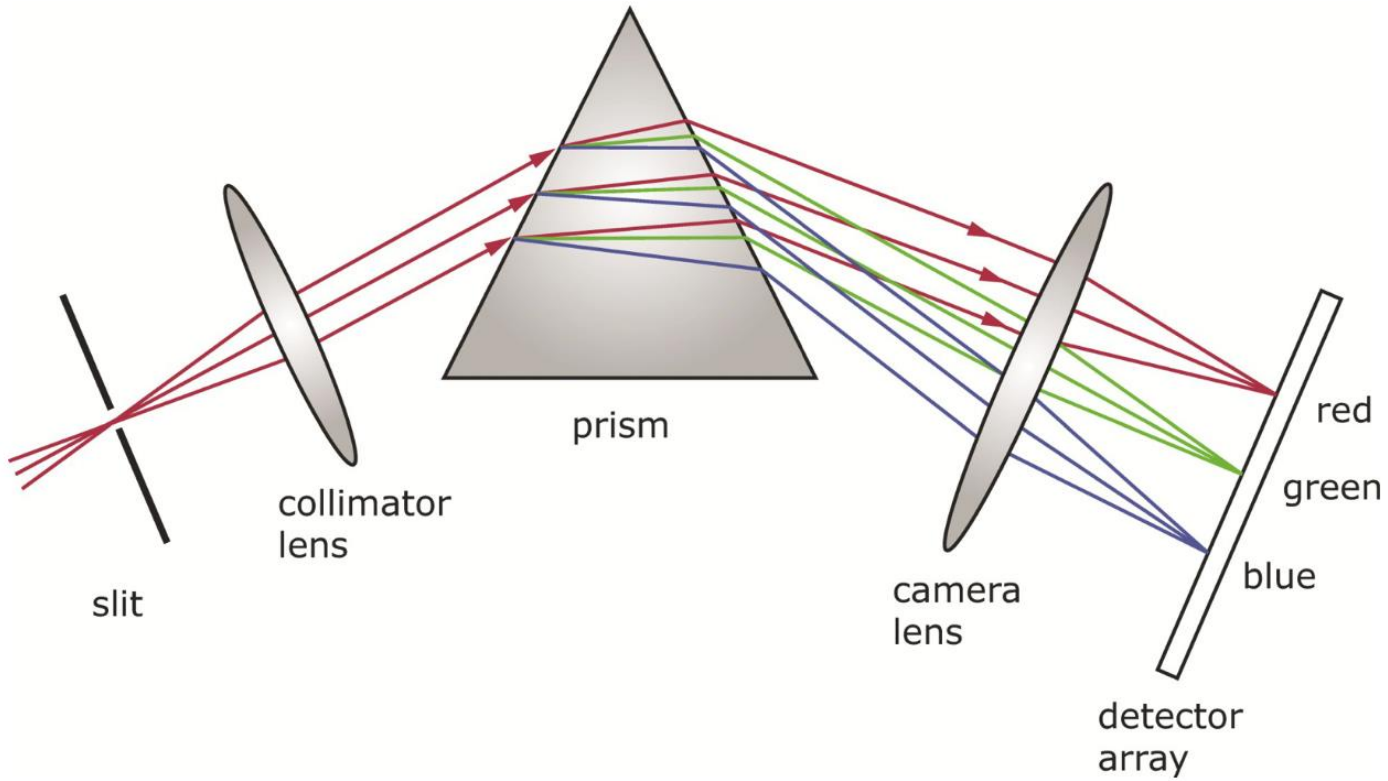
$$\frac{d\theta}{d\lambda} = \frac{B}{a}\frac{dn}{d\lambda}$$

- Thicker prisms have higher dispersion power.



Derivation in Chapter 12 in Obs Book

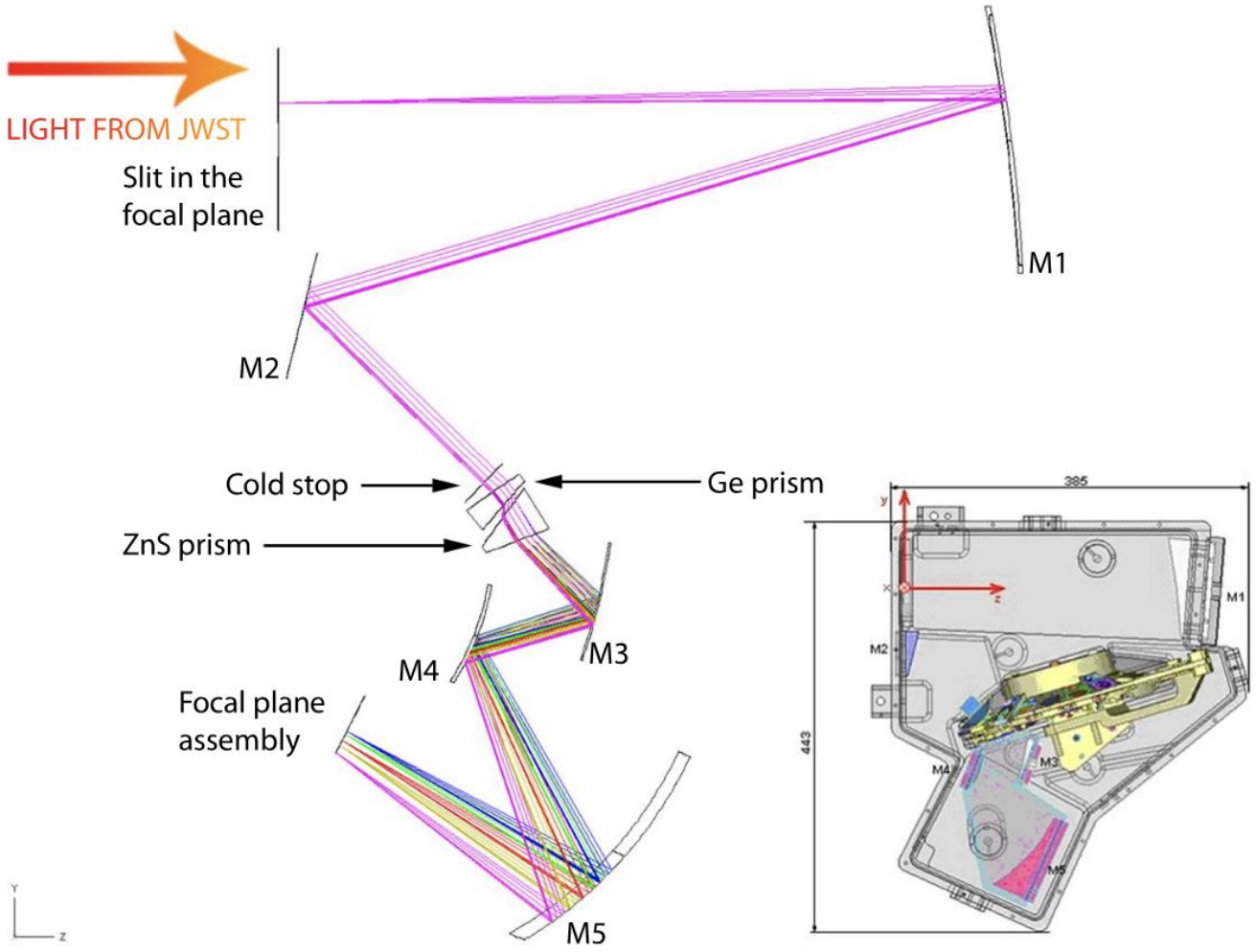
Types of Dispersive Elements: Prisms



If index of refraction n(λ) changes with wavelength, so does the dispersion power.

$$\frac{d\theta}{d\lambda} = \frac{B}{a}\frac{dn}{d\lambda}$$

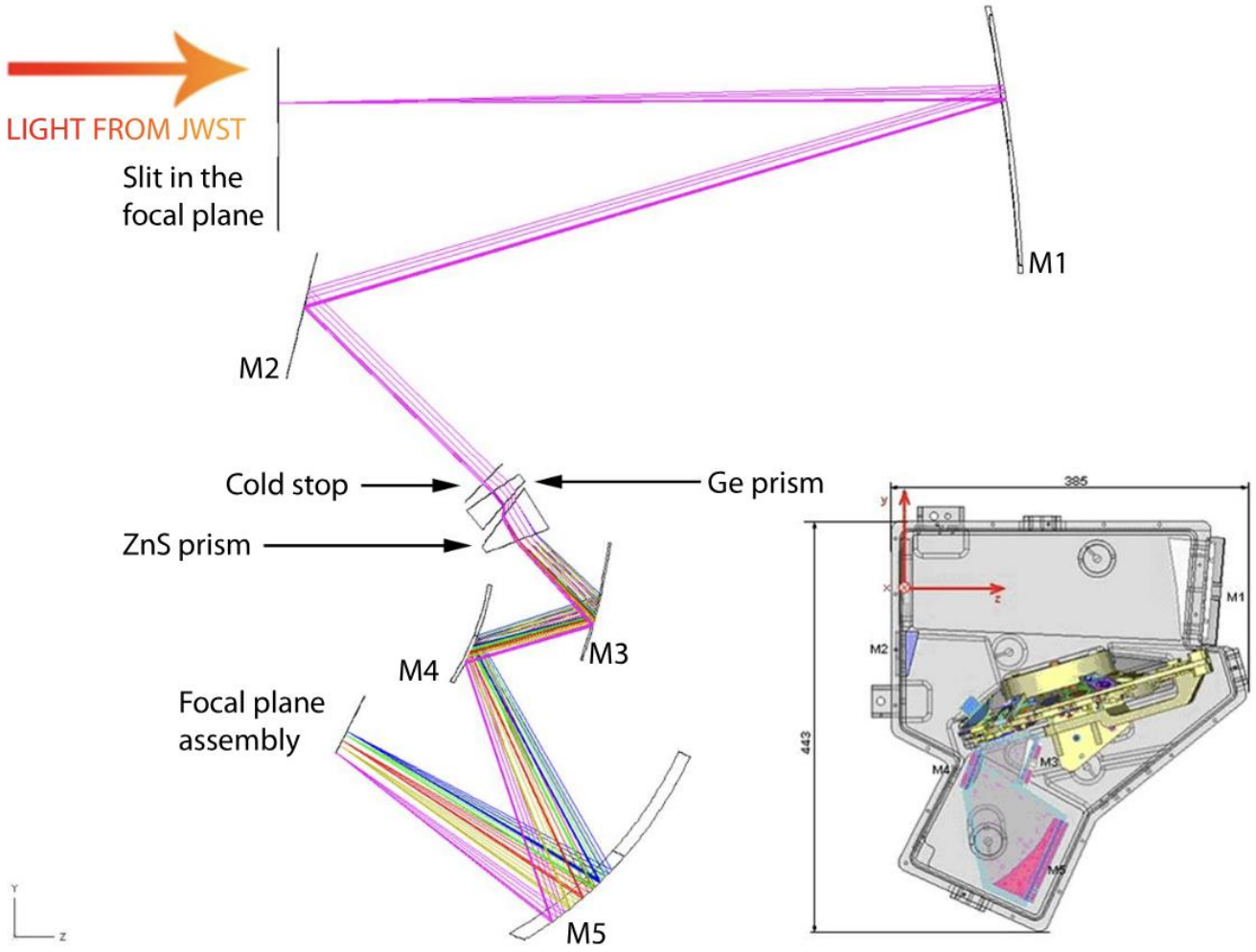
Prism Example: JWST/MIRI



- Prisms can be stacked or placed in series to increase resolution power.
- Several prisms can be used to keep the telescope beam relatively straight.
- This makes it easy to swap between imaging and spectroscopy using the same detector

Fischer+ 2008

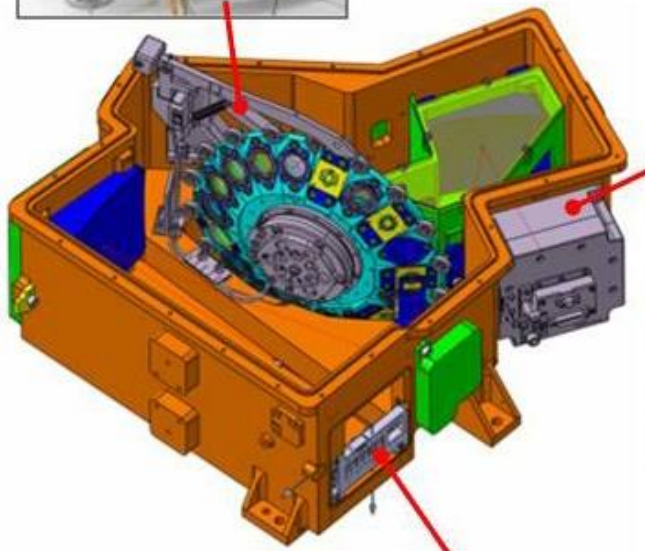
Prism Example: JWST/MIRI





18 Station Filter Wheel

- Filters (10), LRS prism (1), chronographic diaphragms and filters (4), open (1), closed+PAR(1), lens (1)





Focal Plane Module

- 6.7K operating temperature
- 1024 x 1024 Si:As detector array thermally and electrically isolated from housing

Coronagraphic masks

- 4QPM (3)
- Lyot mask (1)



Fischer+ 2008

Jdocs Website (Arizona team provided)

Prism Example: JWST/MIRI

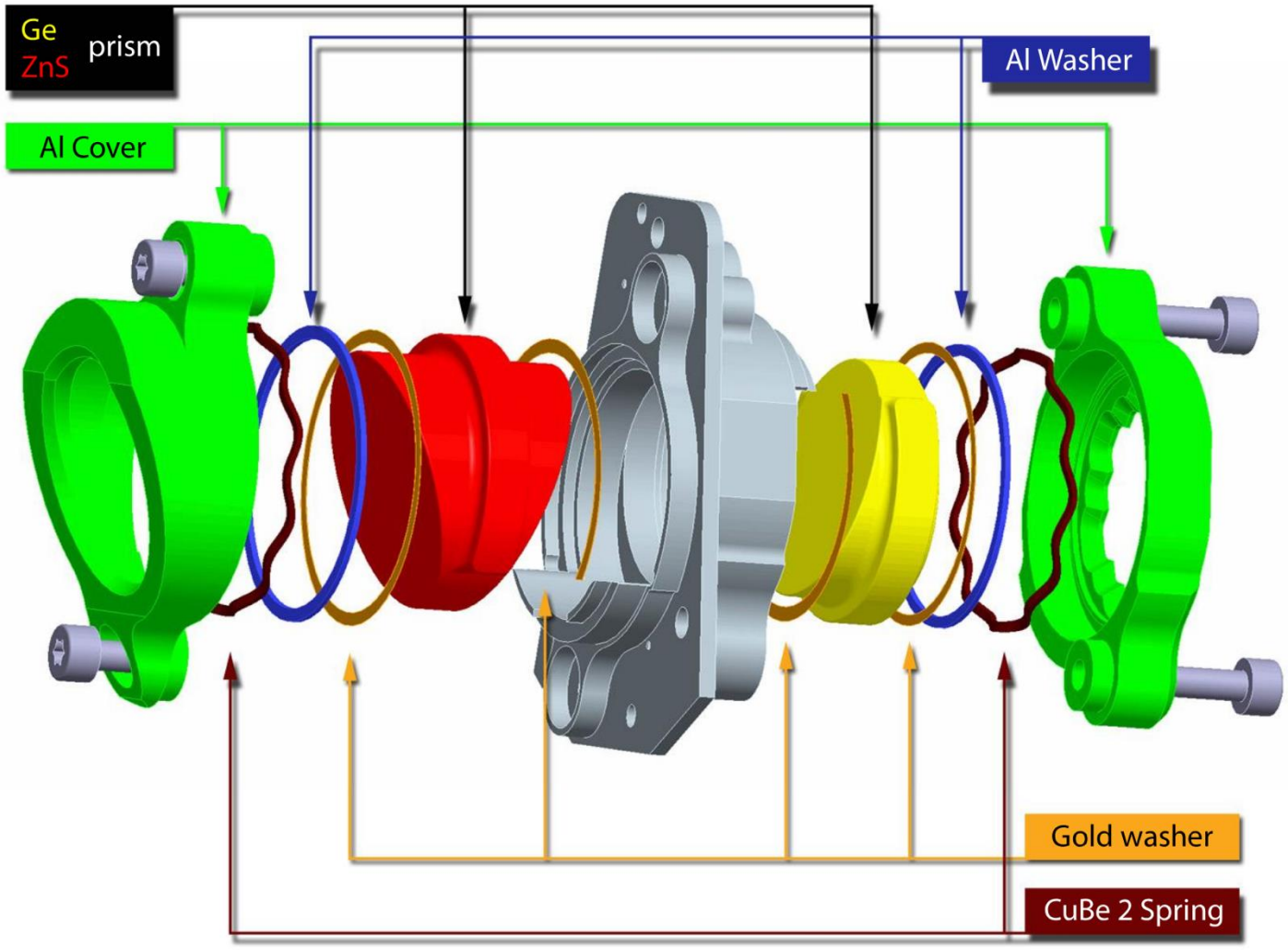


Figure 3. An exploded view of the LRSDPA and its components.

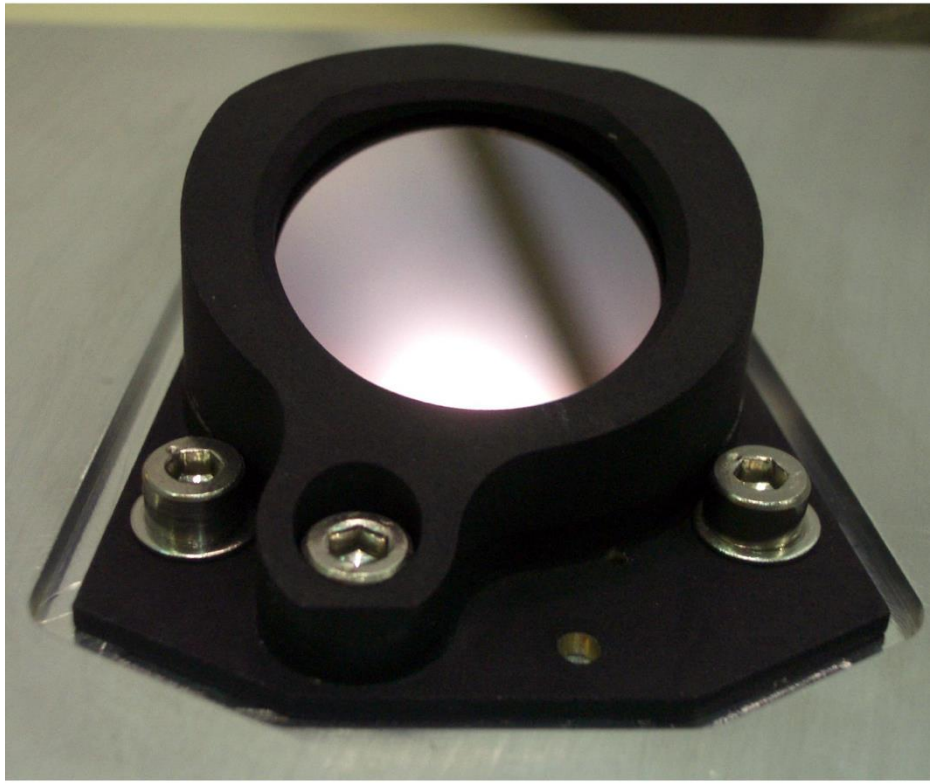


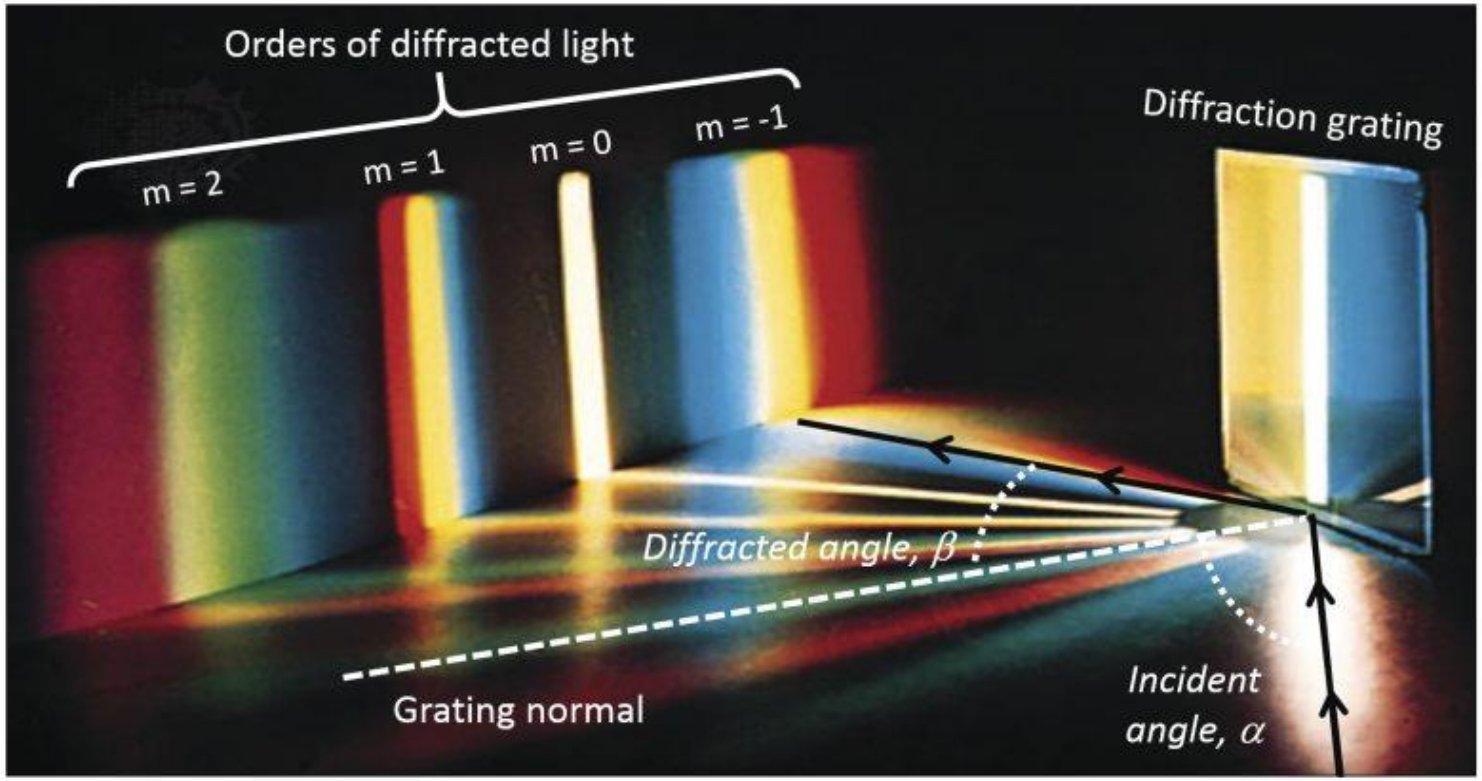
Figure 7. The flight model of the LRSDPA (view onto the ZnS side).

Fischer+ 2008

Types of Dispersive Elements: Gratings

- **Diffraction Gratings** disperse light into separate wavelengths using diffraction. Multiple grooves are used to increase dispersion power.

- Gratings can be reflective or transmissive.

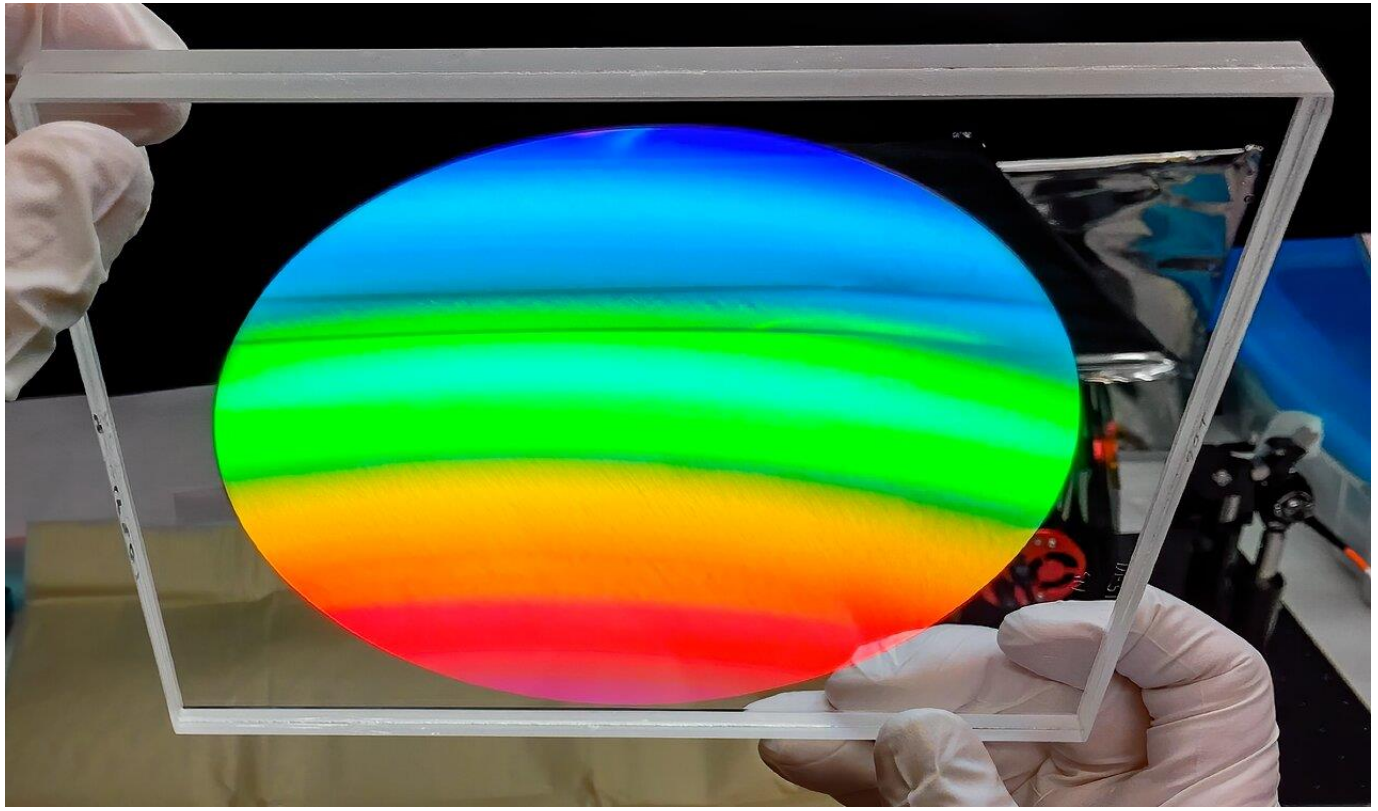


Reflective Grating: Newport

Types of Dispersive Elements: Gratings

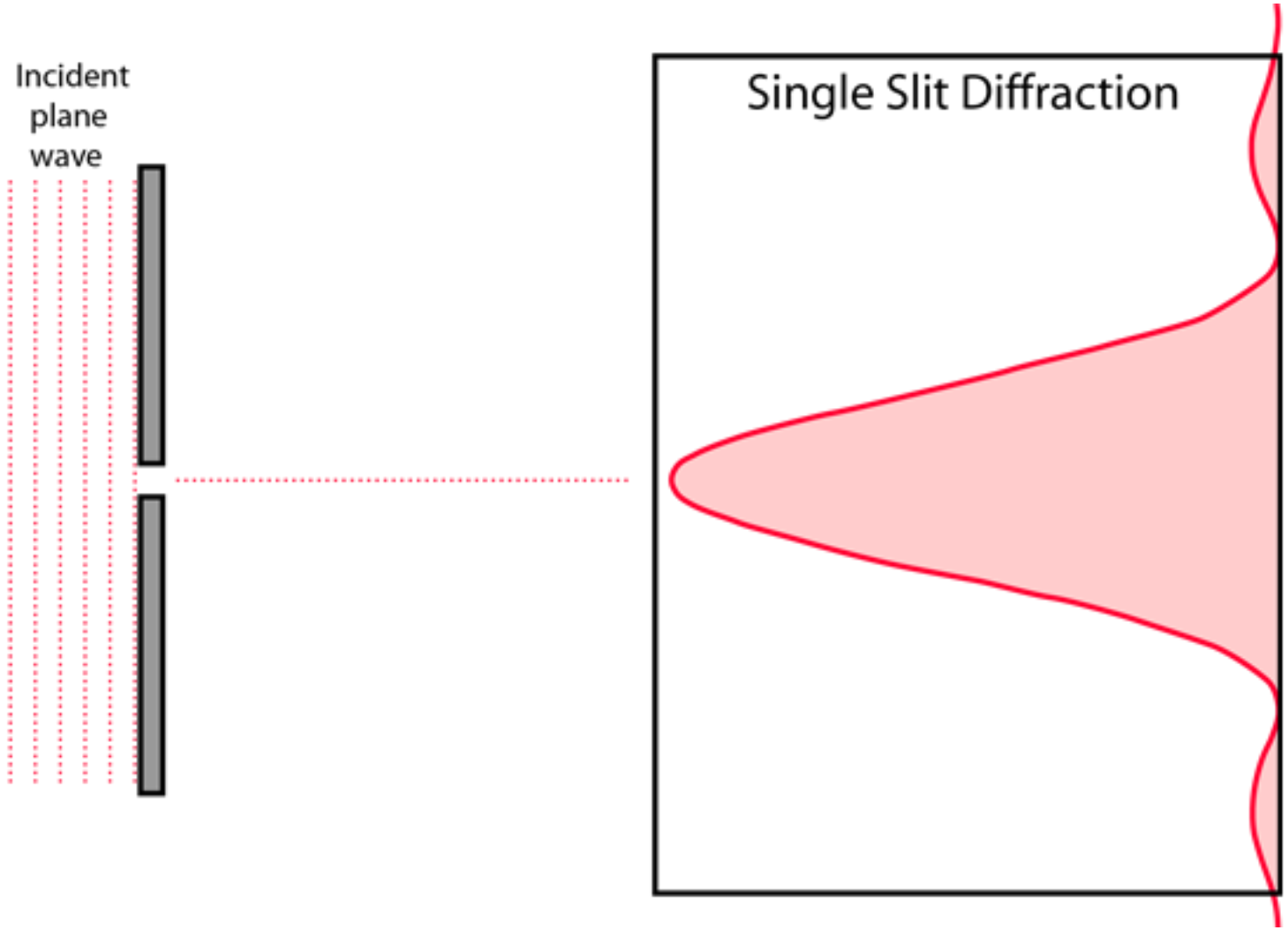
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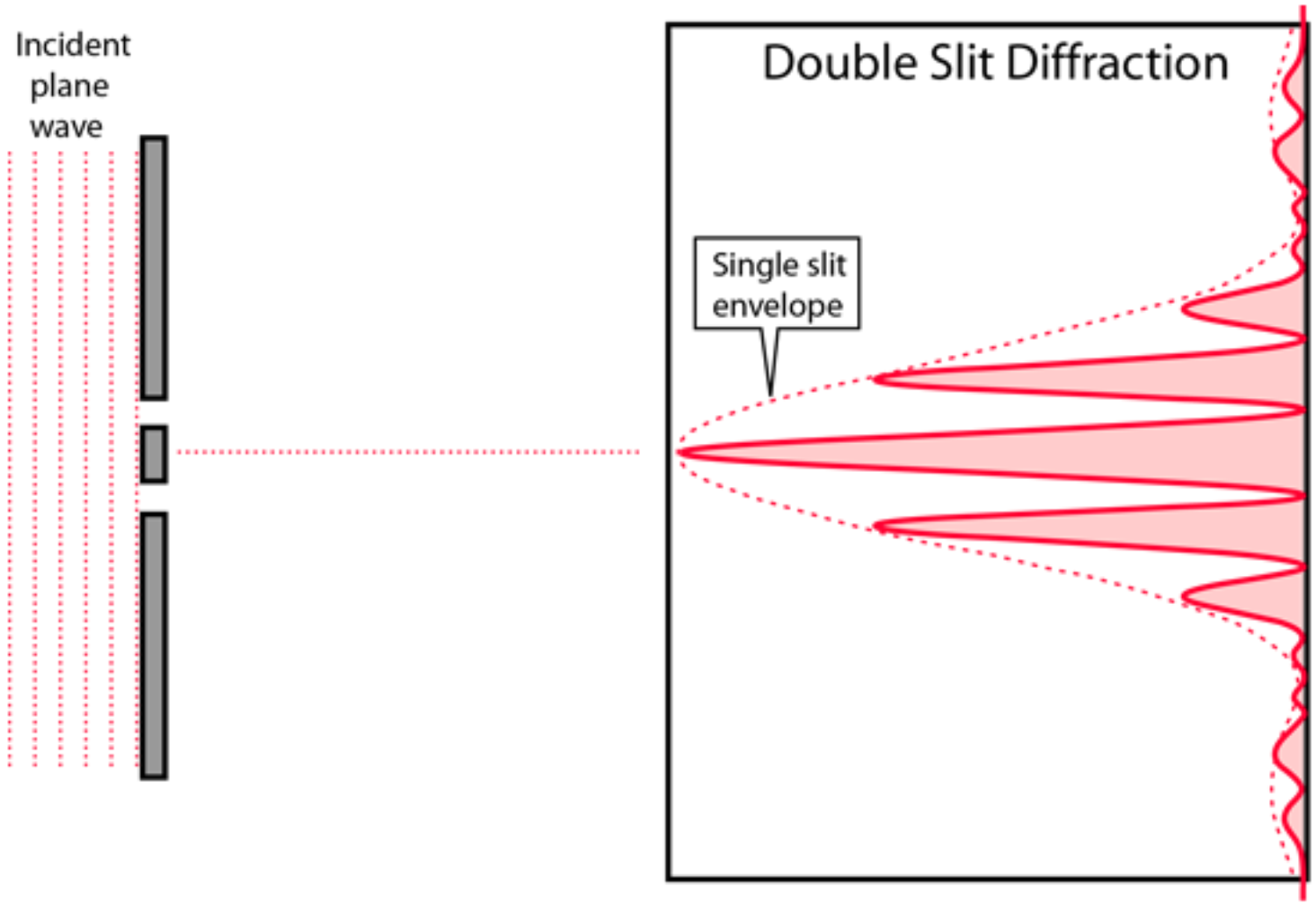
Transmissive Grating: VPH, DESI (NOIRLab)

Monochromatic Diffraction



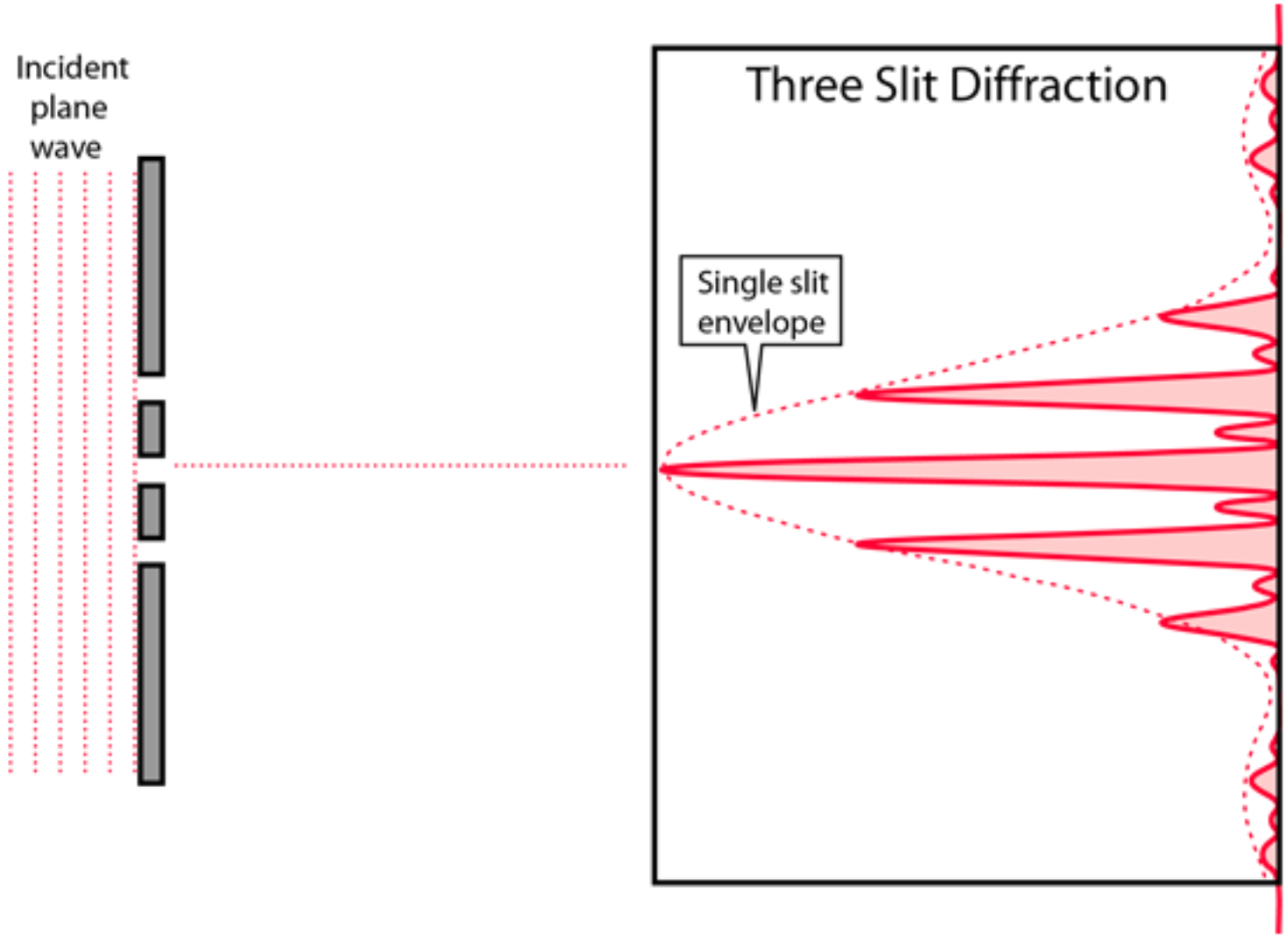
Hyperphysics

Monochromatic Diffraction



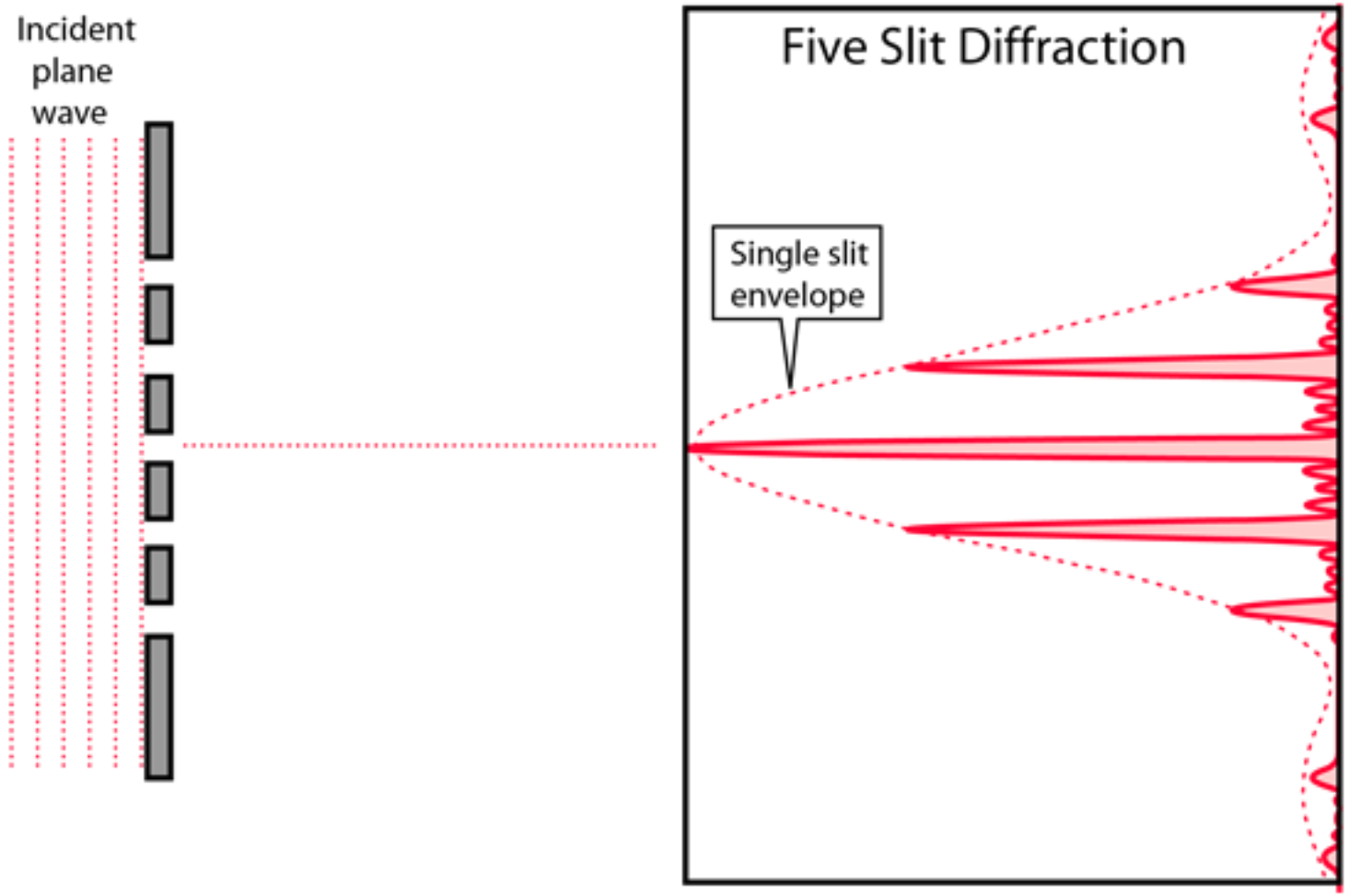
Hyperphysics

Monochromatic Diffraction



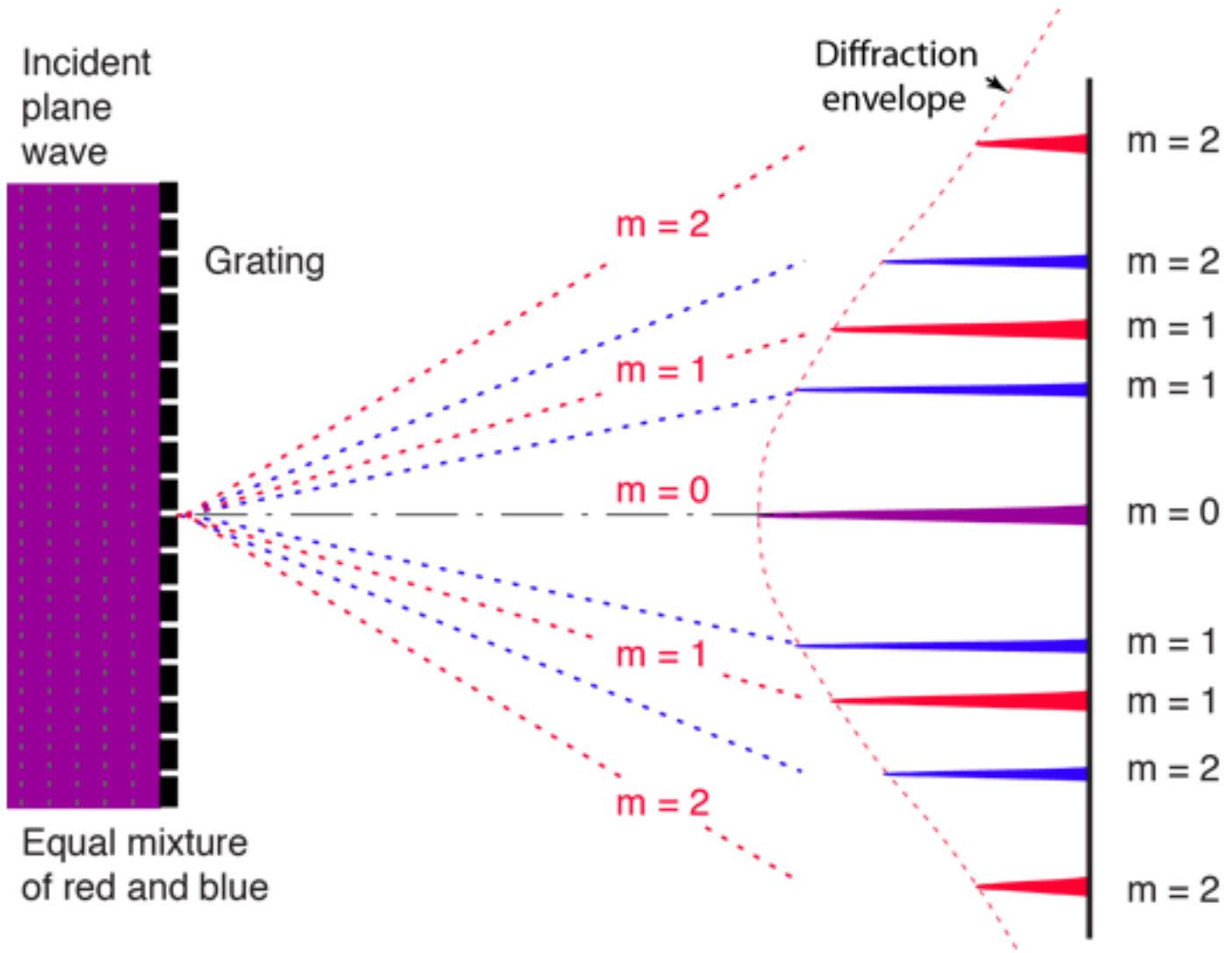
Hyperphysics

Monochromatic Diffraction



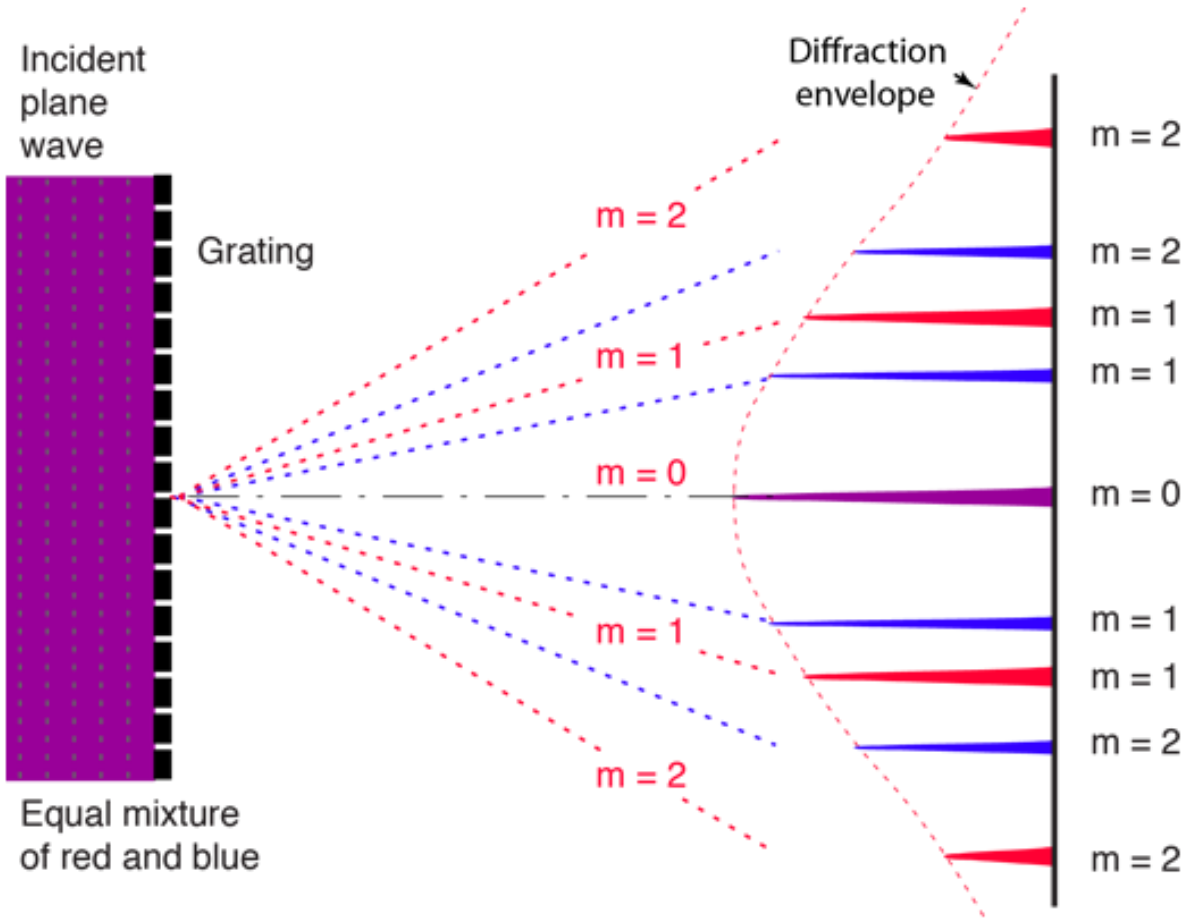
Hyperphysics

Red + Blue Diffraction

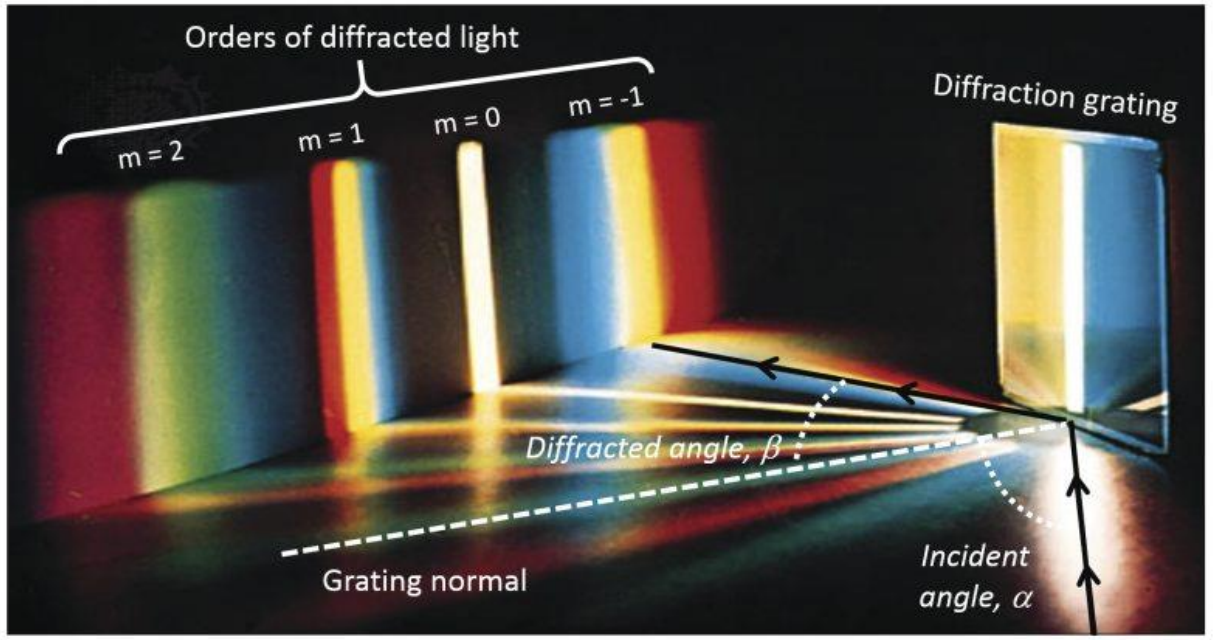


Hyperphysics

Red + Blue Diffraction

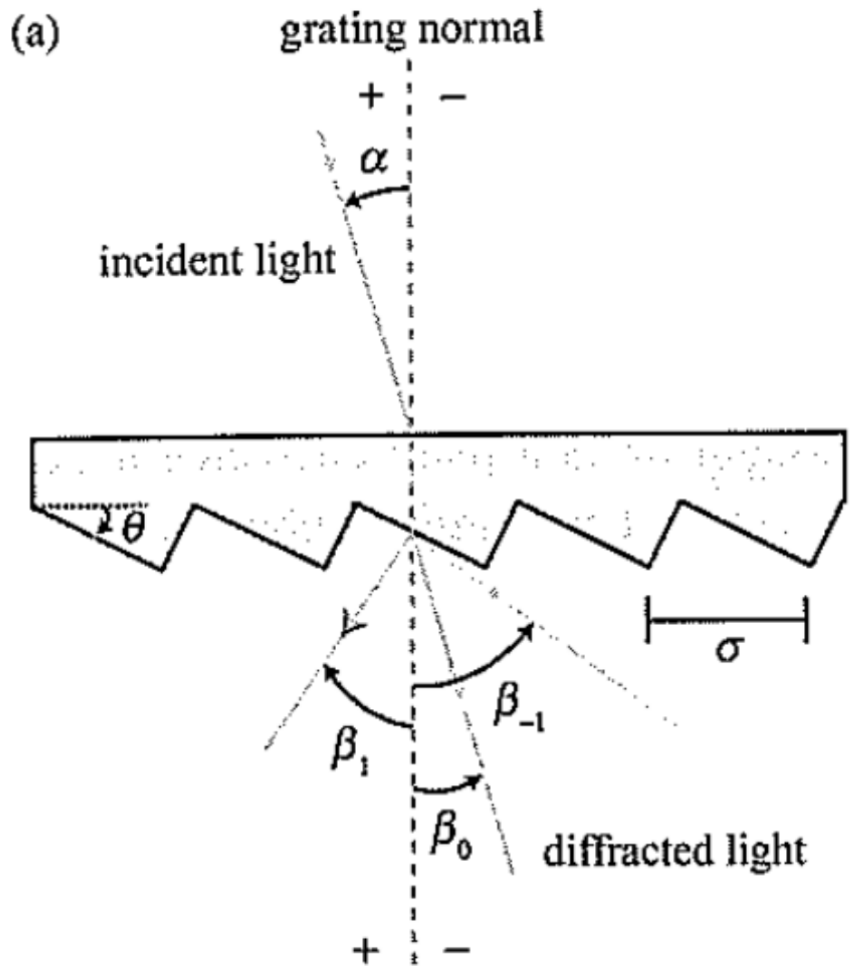


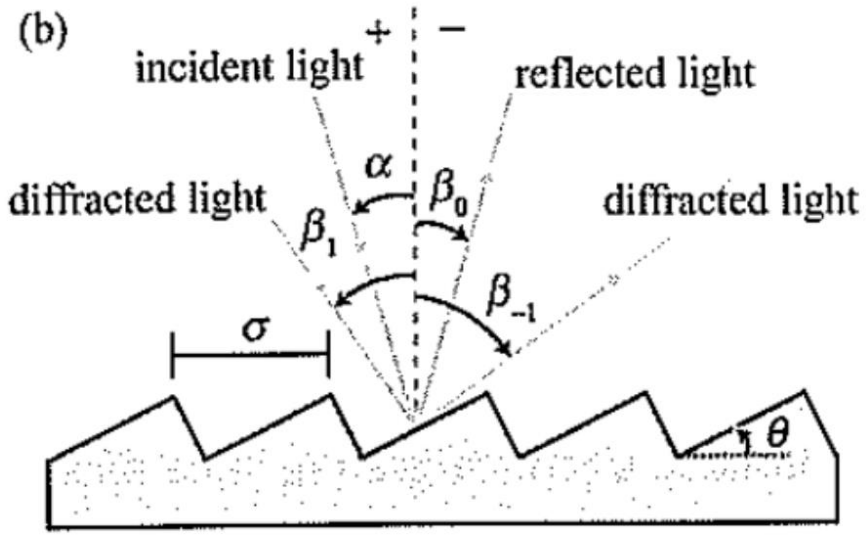
Hyperphysics



Reflective Grating: Newport

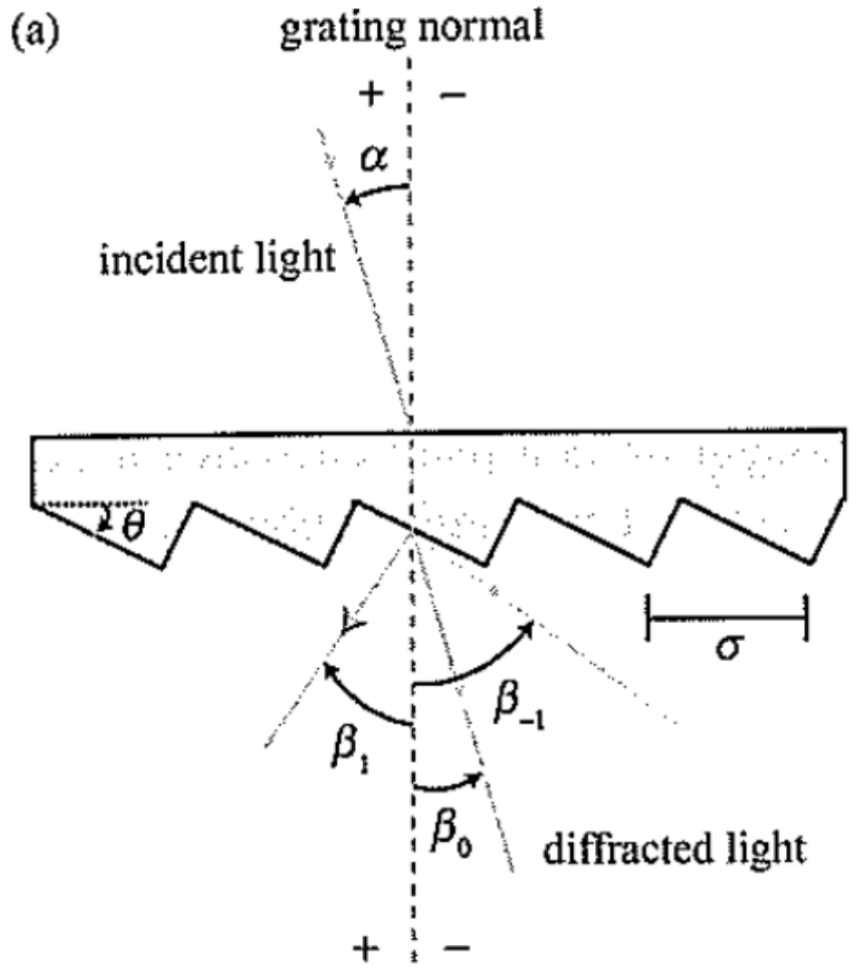
Grating Parameters





sigma=groove spacing
theta=blaze angle
alpha=incident angle
beta=angle of diffraction

The Grating Equation



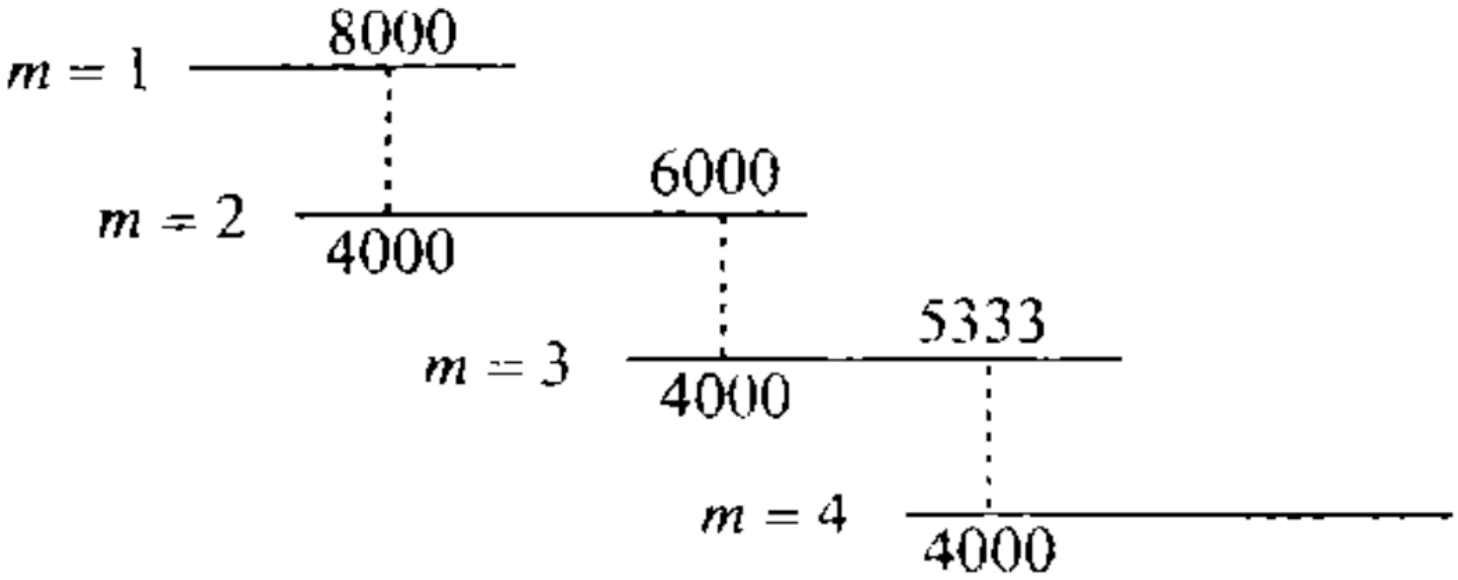
The **Grating Equation** is used to describe how light of a specific wavelength (λ) will be diffracted at different orders(m). Orders can only be integer values.

Groove or line spacing (σ) and order determines the grating's dispersive power.

$$m\lambda = \sigma(\sin\alpha + \sin\beta)$$

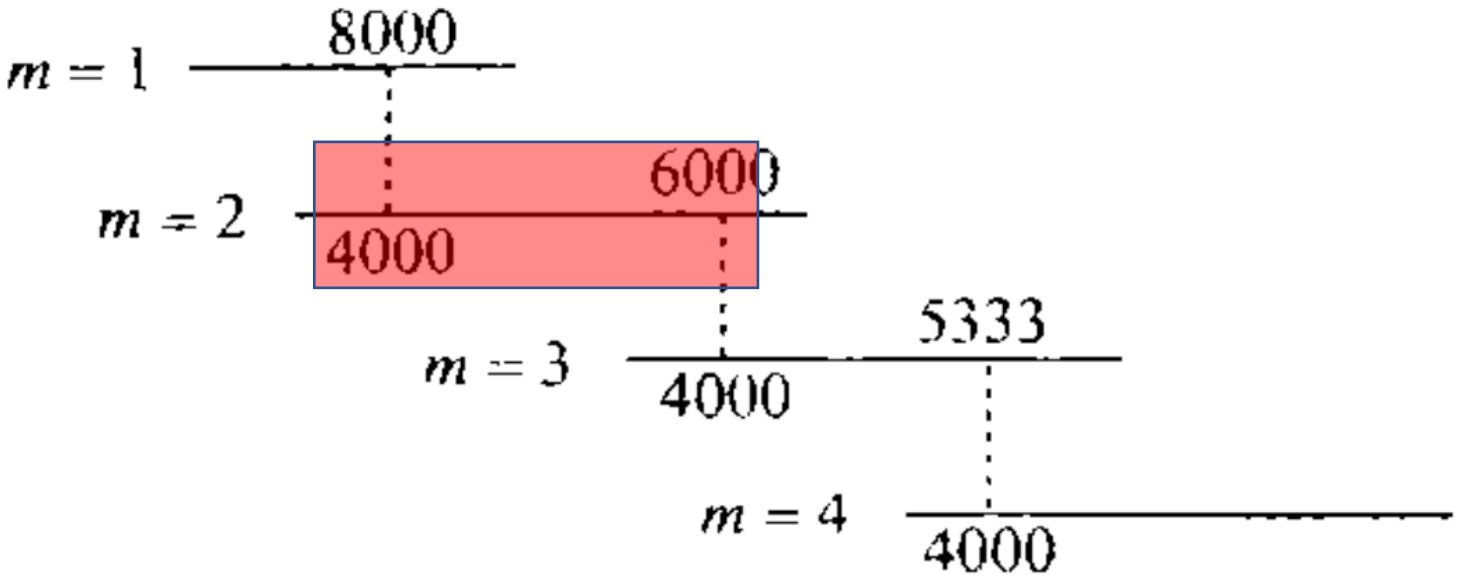
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Free Spectral Range



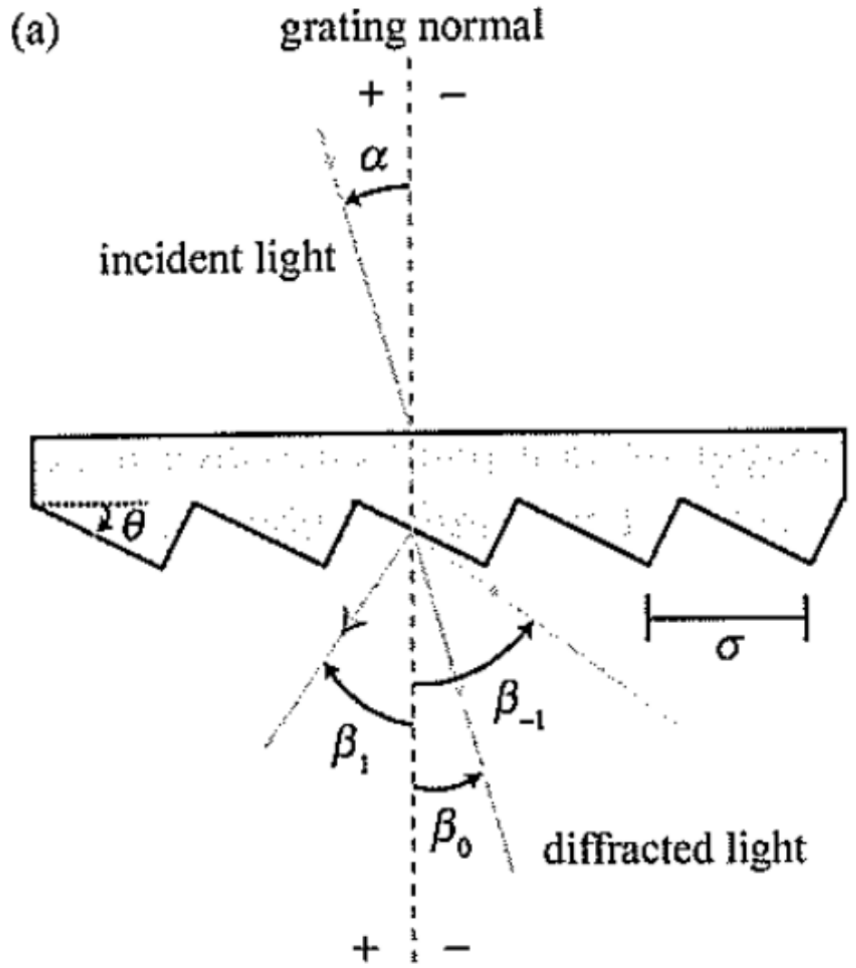
Spectral orders can often overlap, therefore its important to isolate the relevant wavelength range with filters or another mechanism.

Free Spectral Range



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Gratings: Echelle vs Echellette

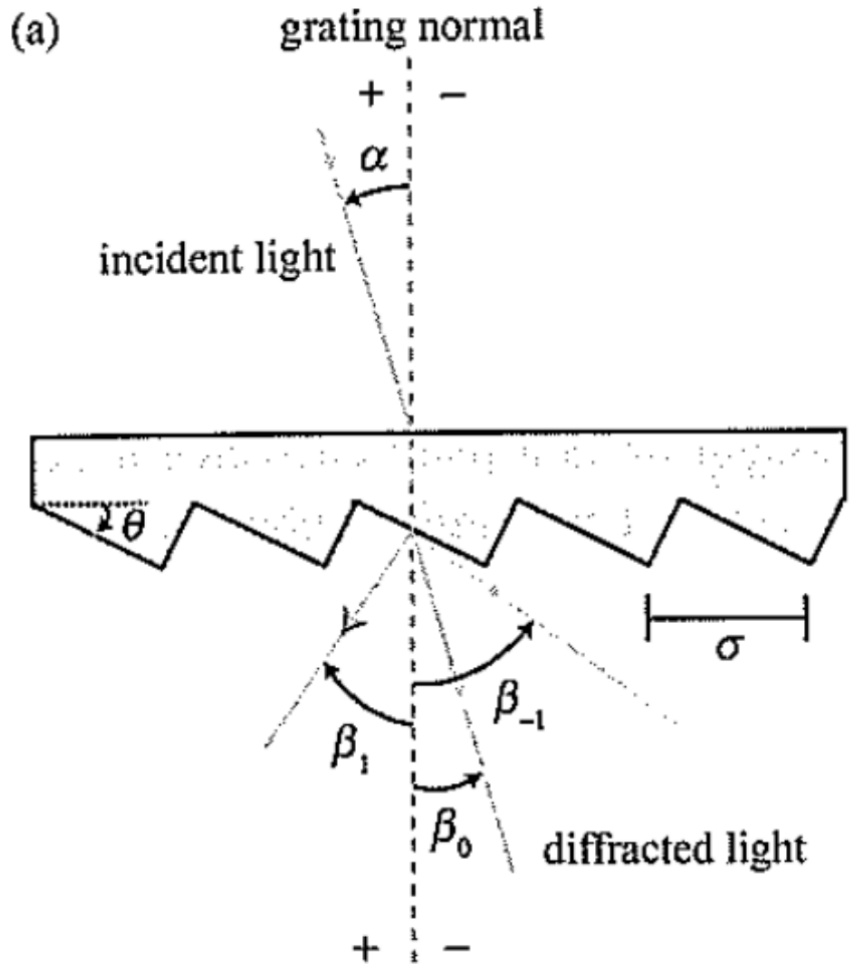


Echellette gratings tend to use small groove spacings ($\boldsymbol{\sigma}$, hundreds/mm). Included in instruments at lower incident angles (α) for use at low orders.

$$\boldsymbol{m\lambda = \sigma\,(\sin\alpha + \sin\beta)}$$

.

Gratings: Echelle vs Echellette



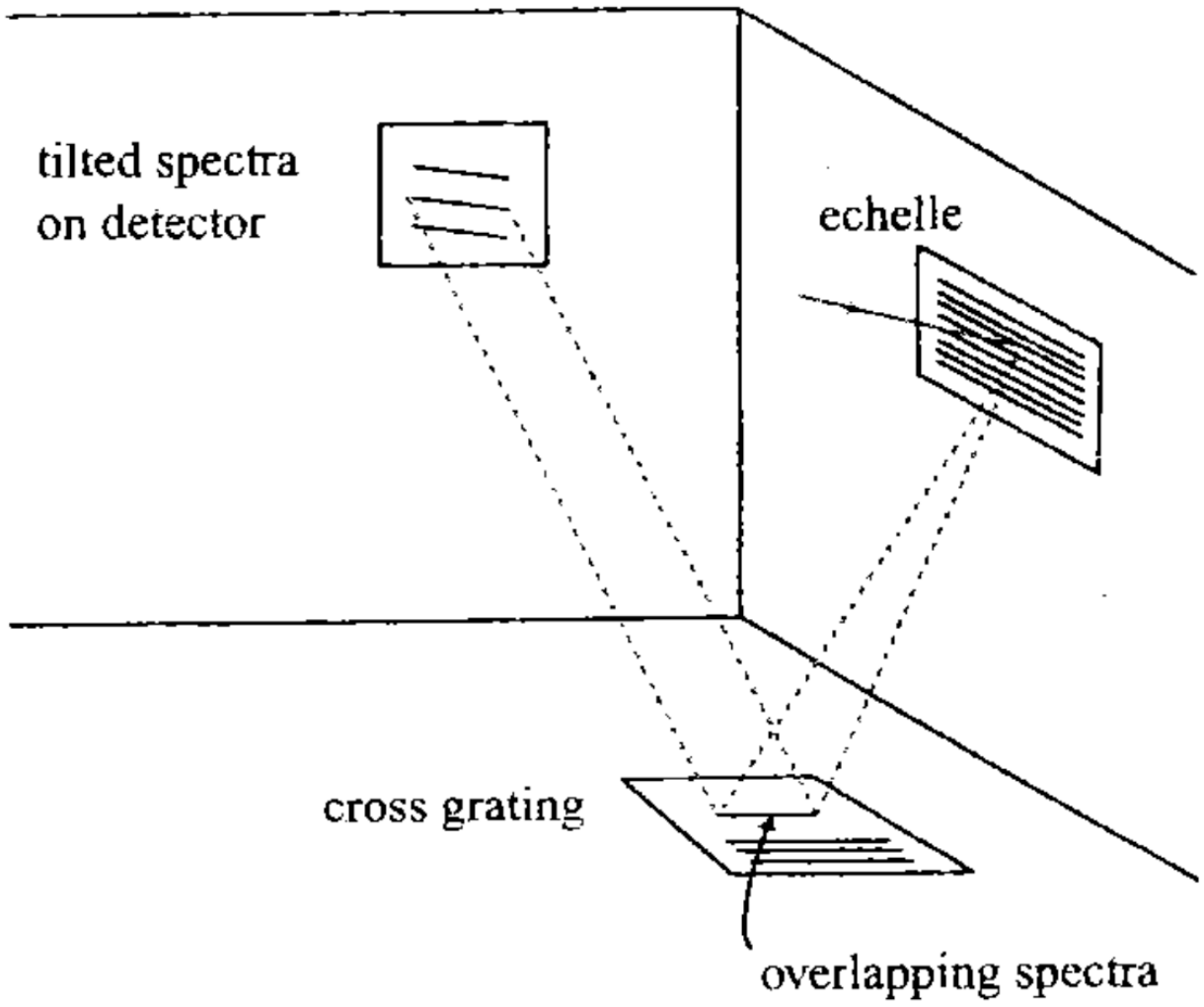
Echelle gratings tend to use larger groove spacings ($\boldsymbol{\sigma}$, ten/mm) and steep blaze angles. Included in instruments at high incident angles (α) for use at higher orders.

You usually use another grating or disperser behind echelle gratings to separate overlapping orders.

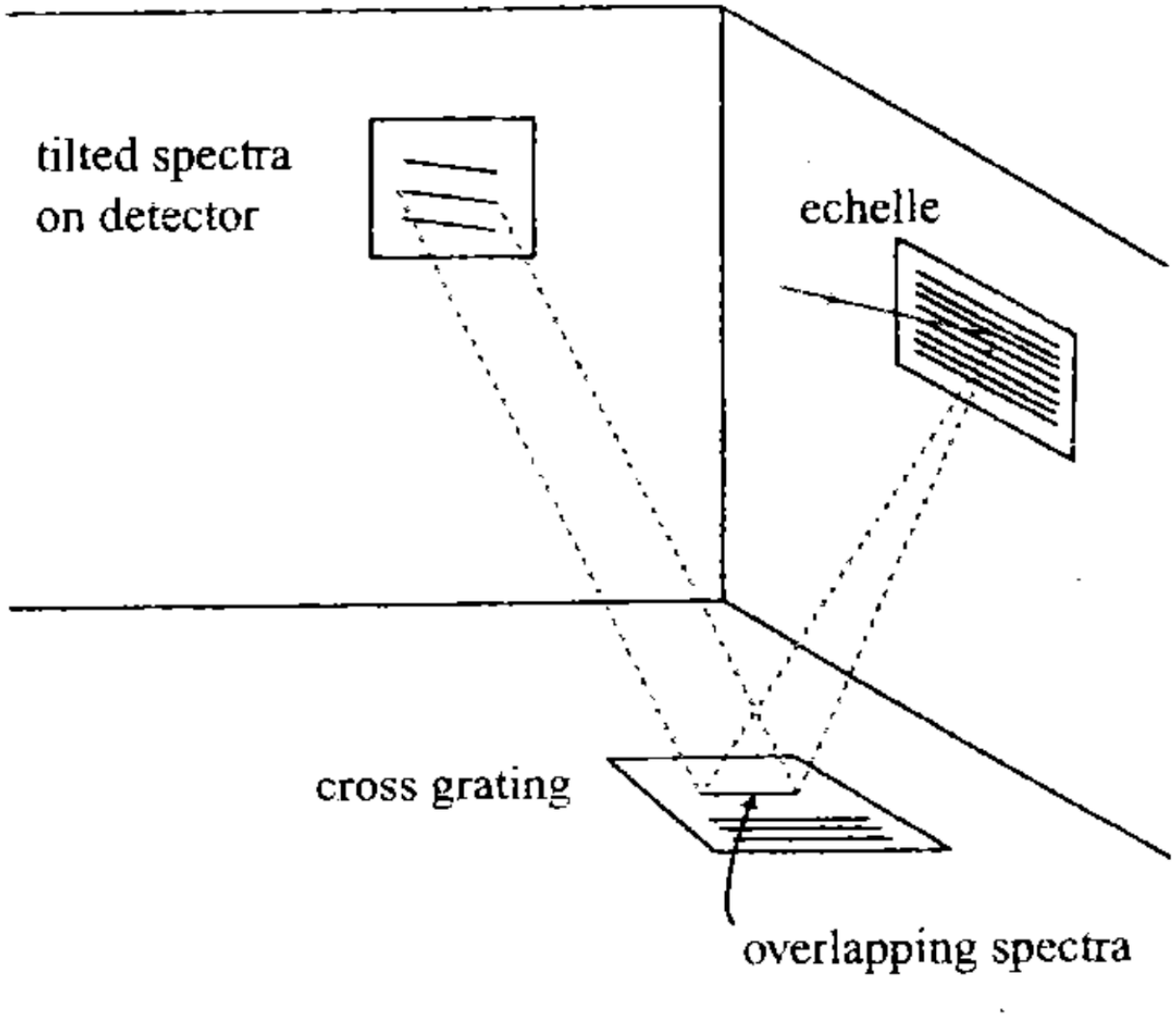
$$\boldsymbol{m\lambda = \sigma\,(\sin\alpha + \sin\beta)}$$

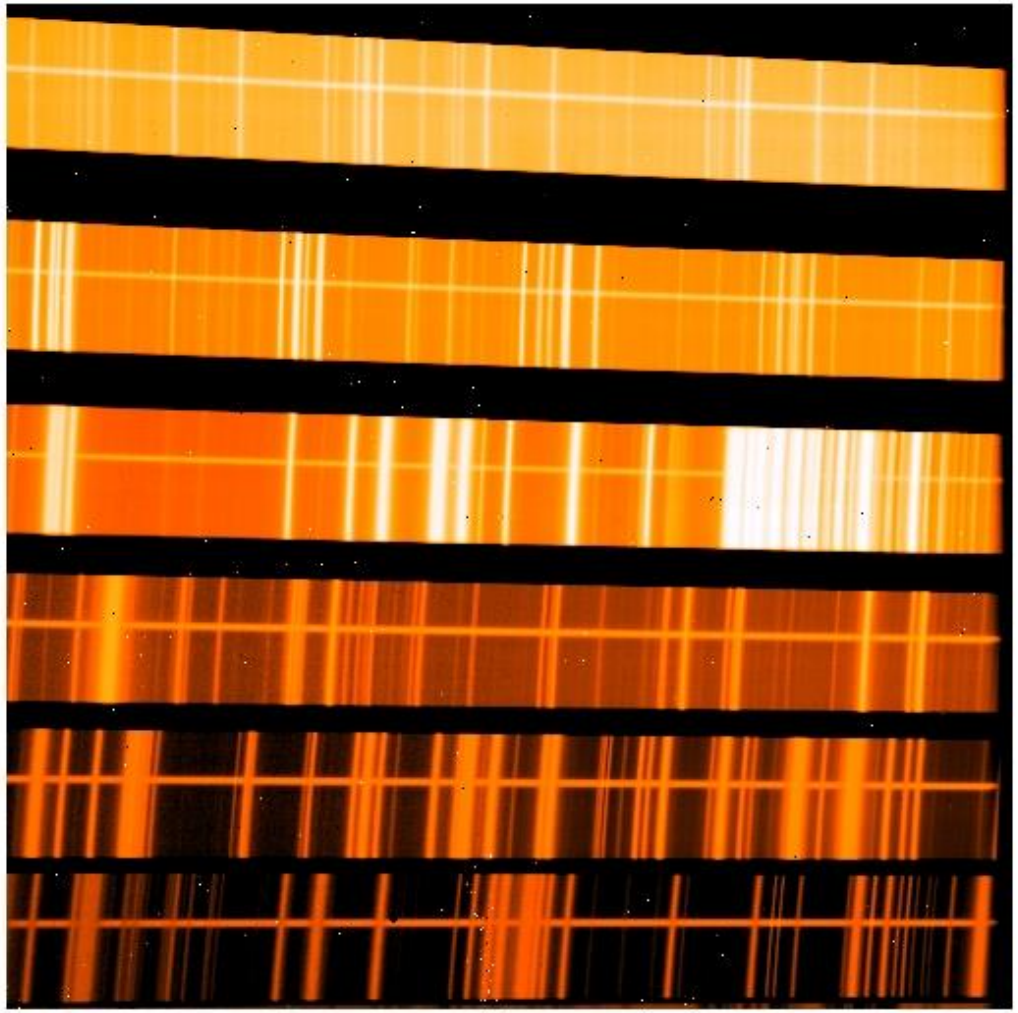
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Gratings: Echelle Grating with Cross Disperser

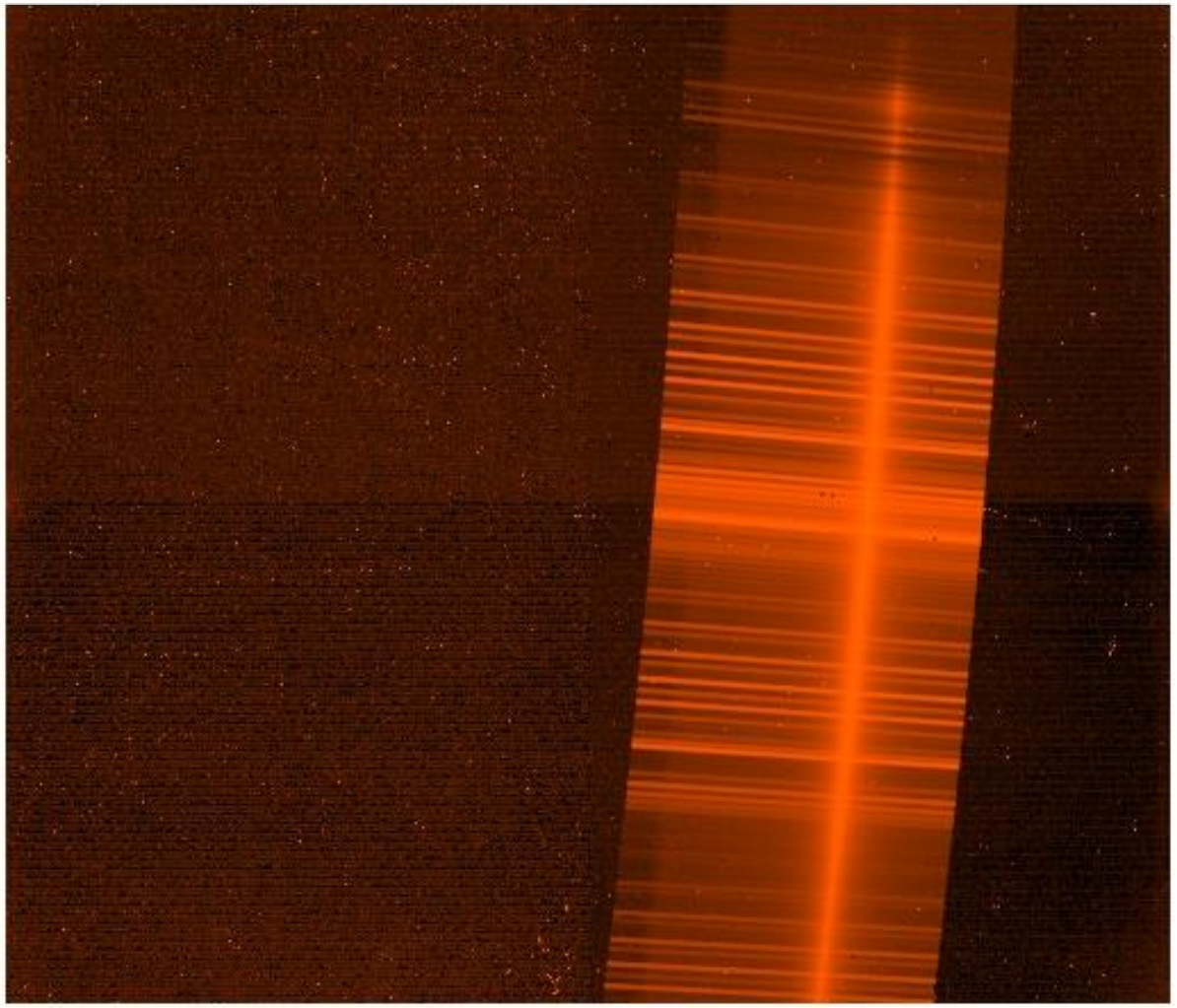


Gratings: Echelle Grating with Cross Disperser

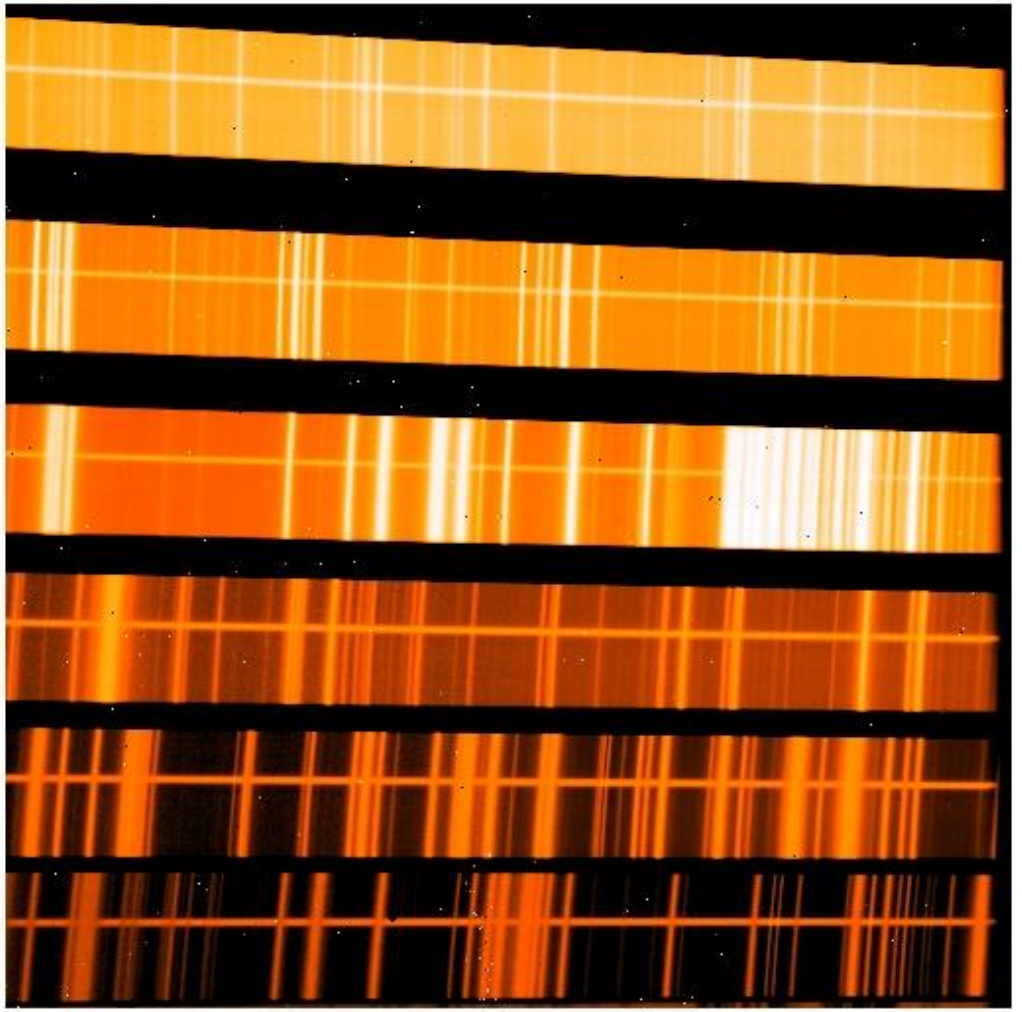




Keck/NIRSpec

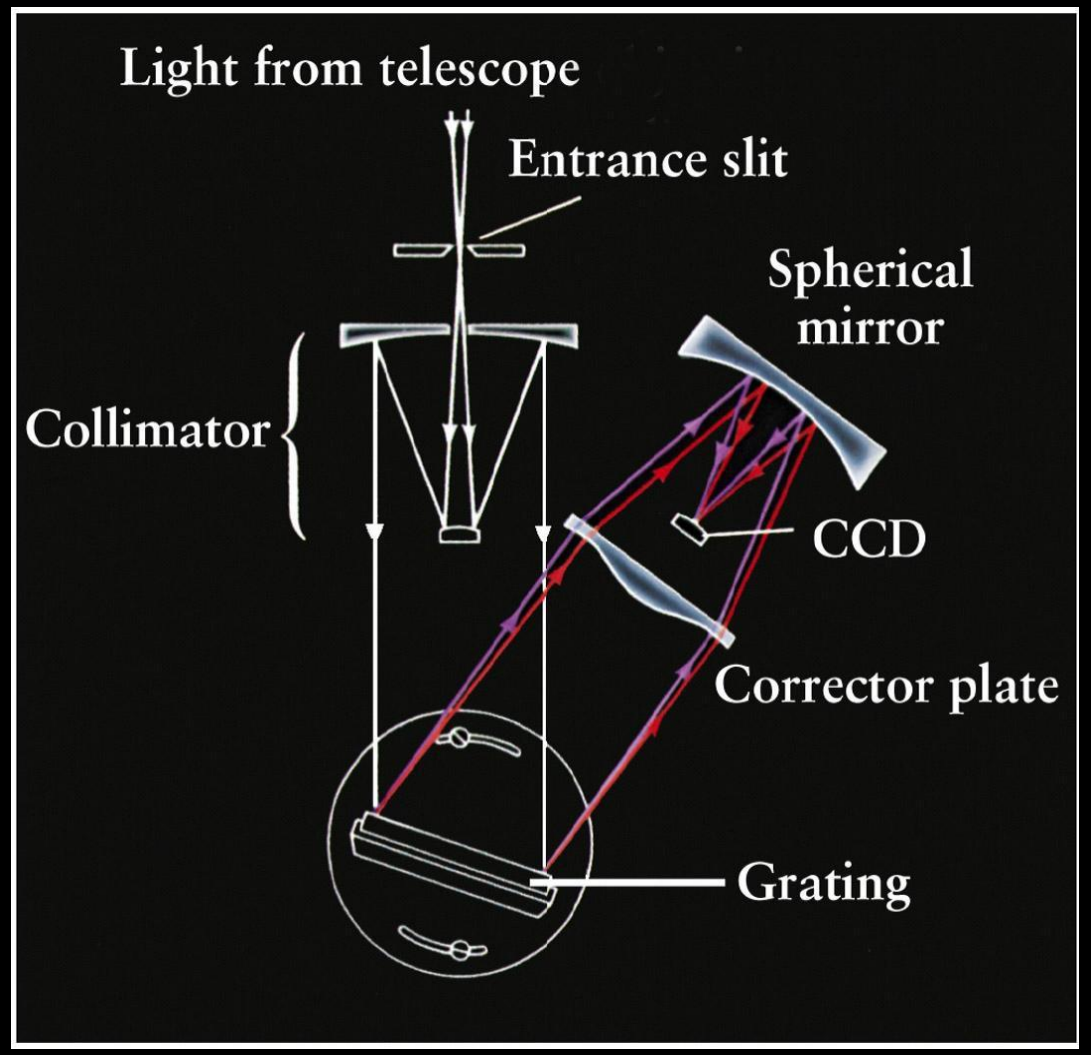


Single Grating (similar to B&C Spectrograph)



Grating + Cross Disperser

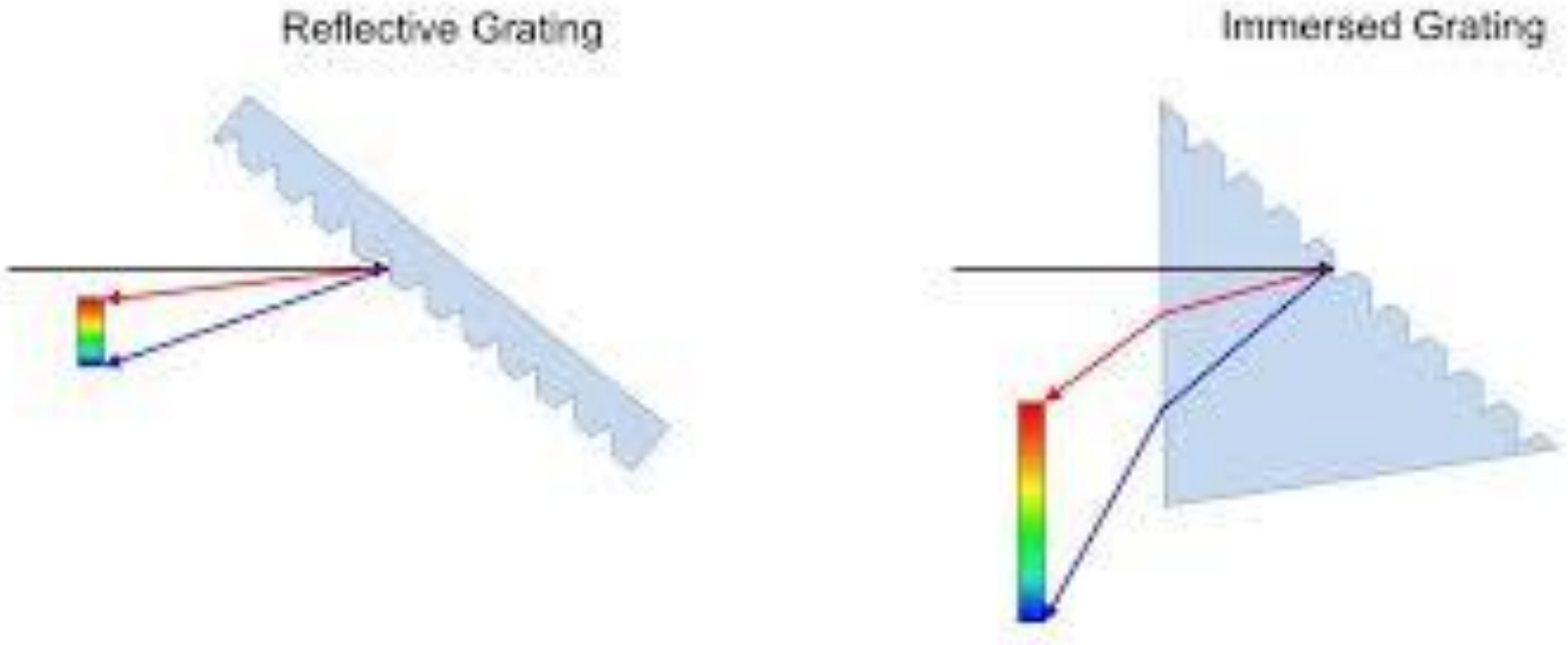
Components of a Spectrograph with Grating



A reflective grating can be rotated to access different orders and wavelengths with your detector.

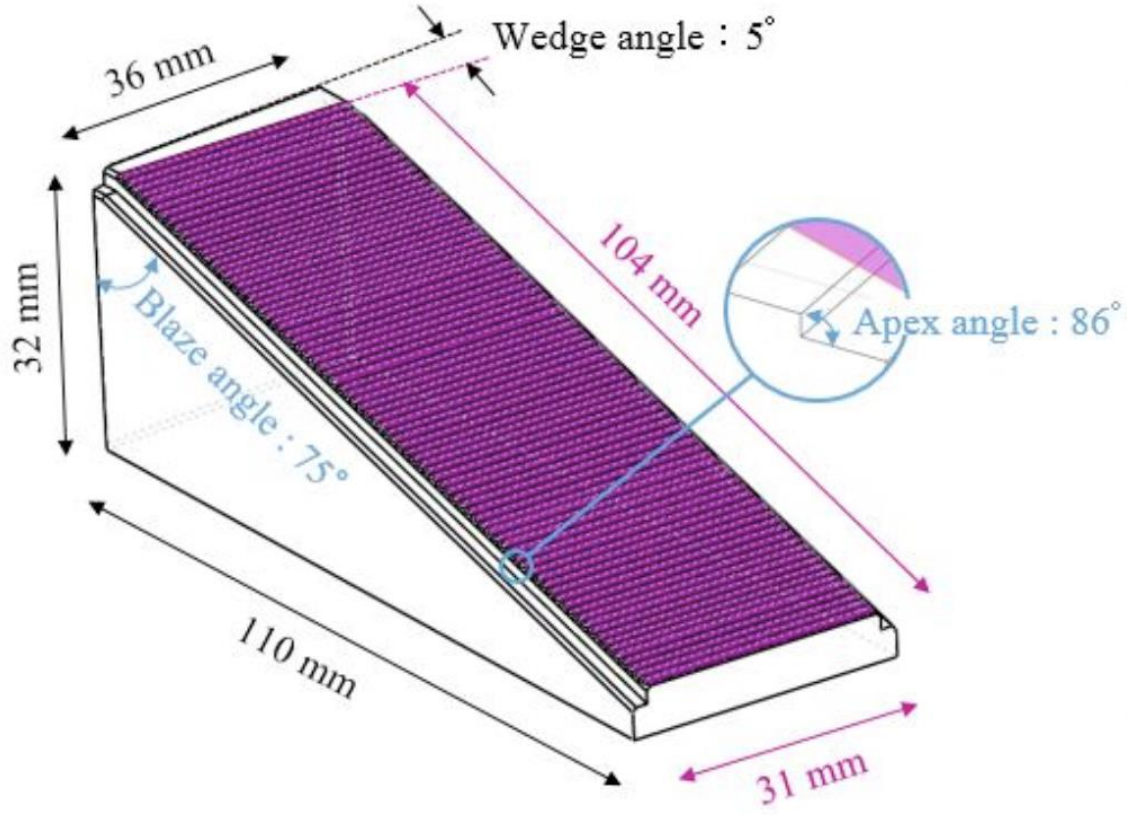
Some instruments keep gratings fixed, but rotate through several to access different wavelengths.

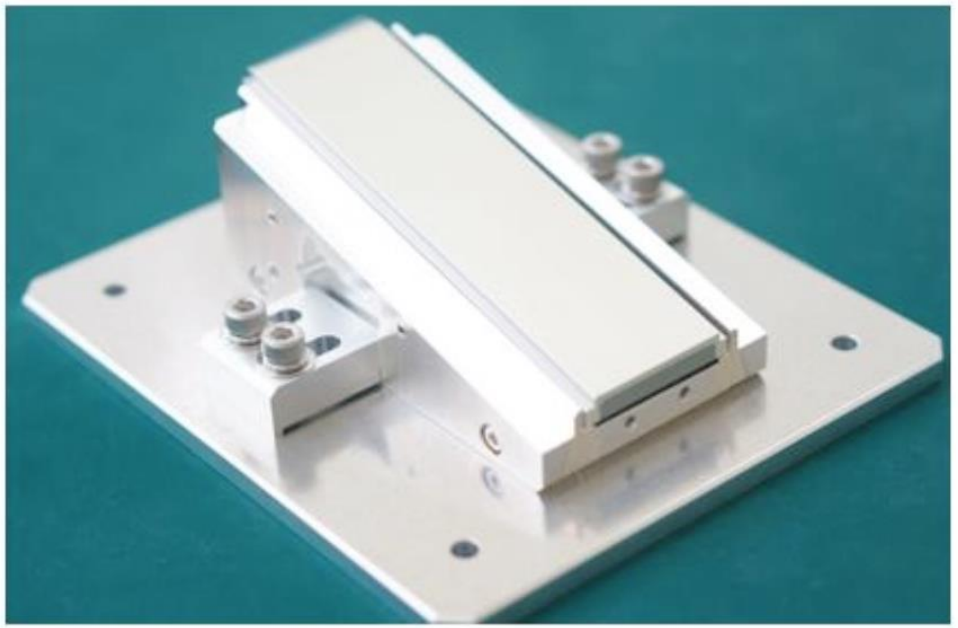
Other Types of Gratings: Immersion Gratings



Immersion gratings are reflective gratings with material of some high index of refraction (n) that can increase the dispersive power of the grating.

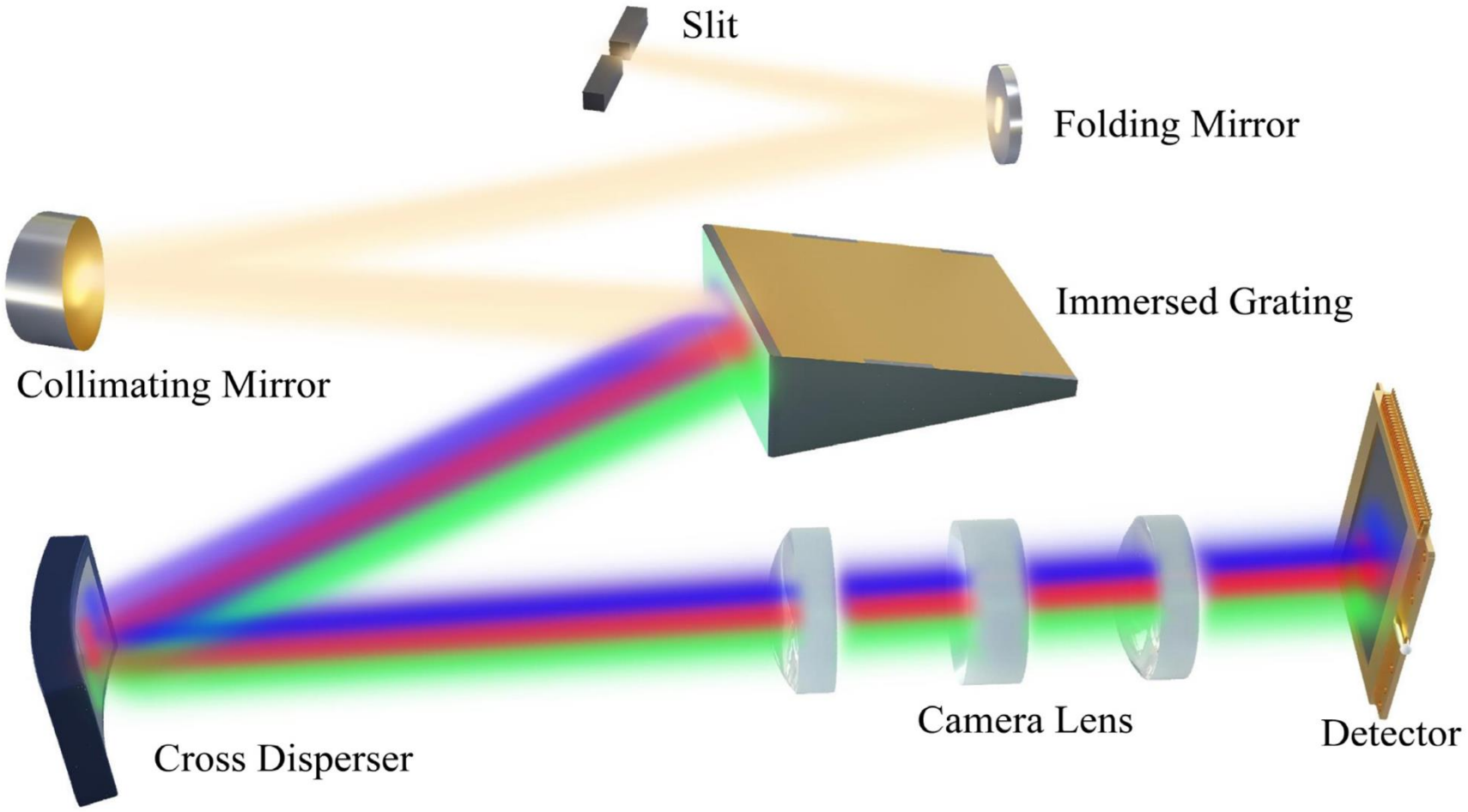
Other Types of Gratings: Immersion Gratings





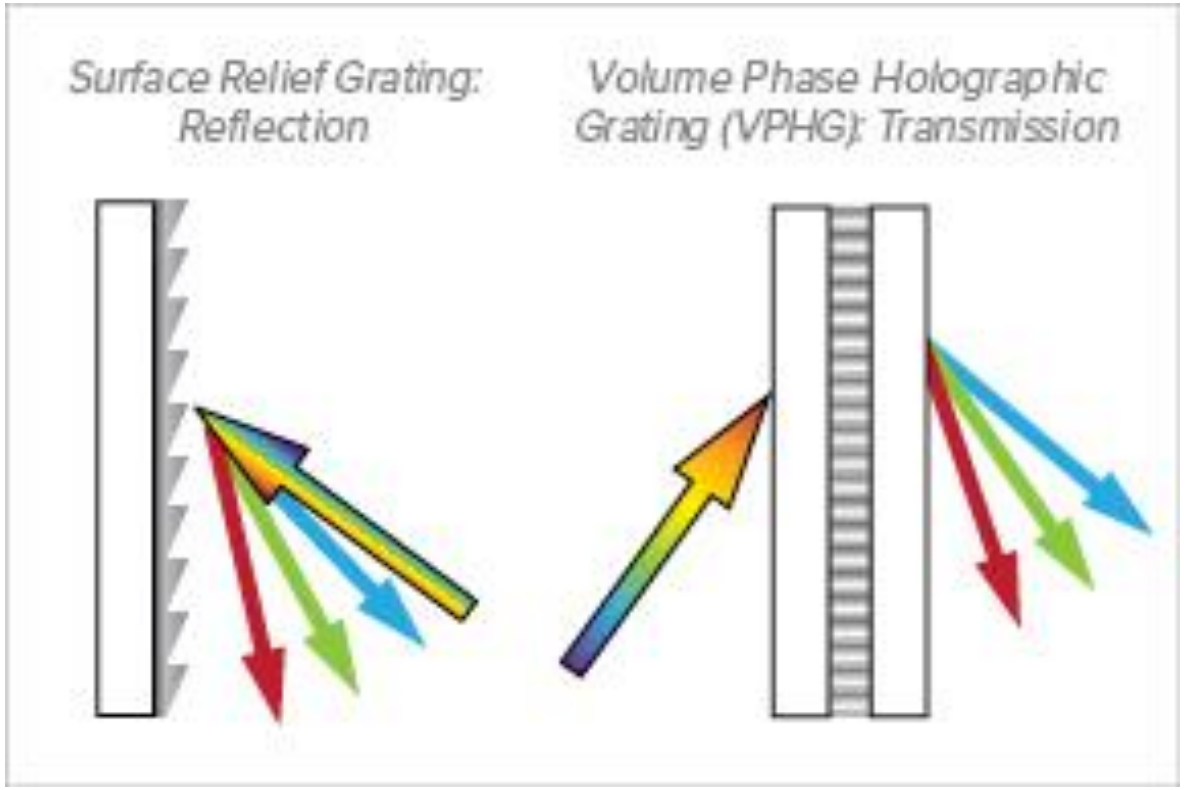
Arasaki+ 2016

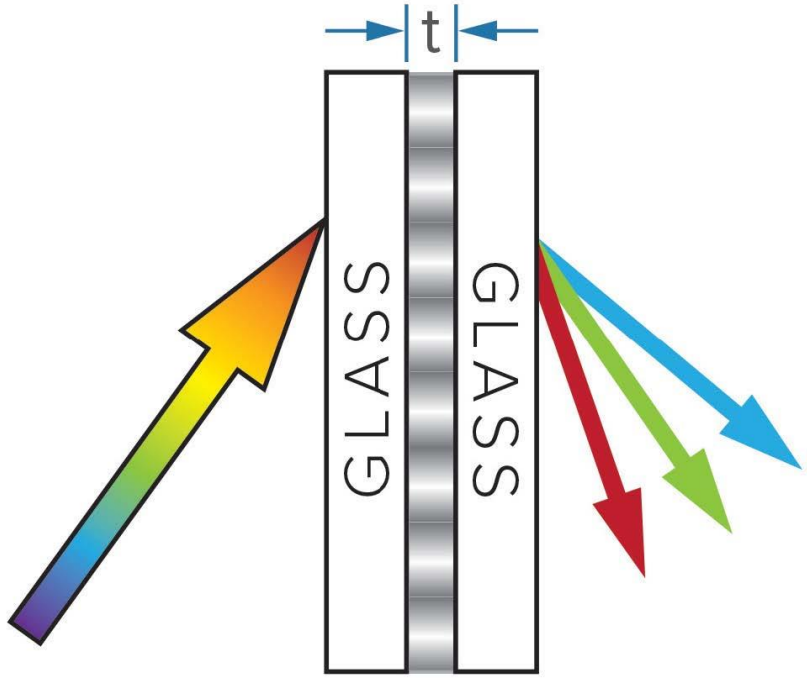
Other Types of Gratings: Immersion Gratings



Wang+ 2022

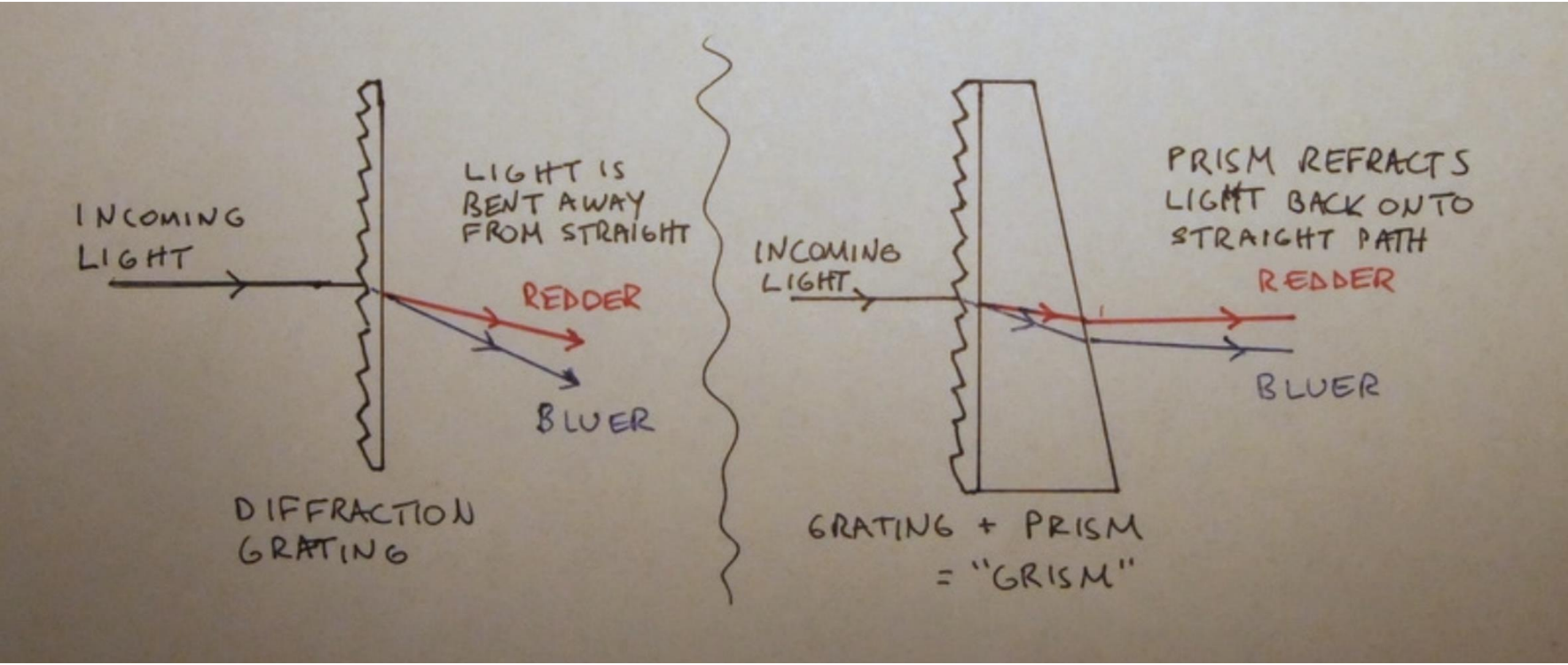
Other Types of Gratings: Volume Phase Holographic (VPH) Gratings





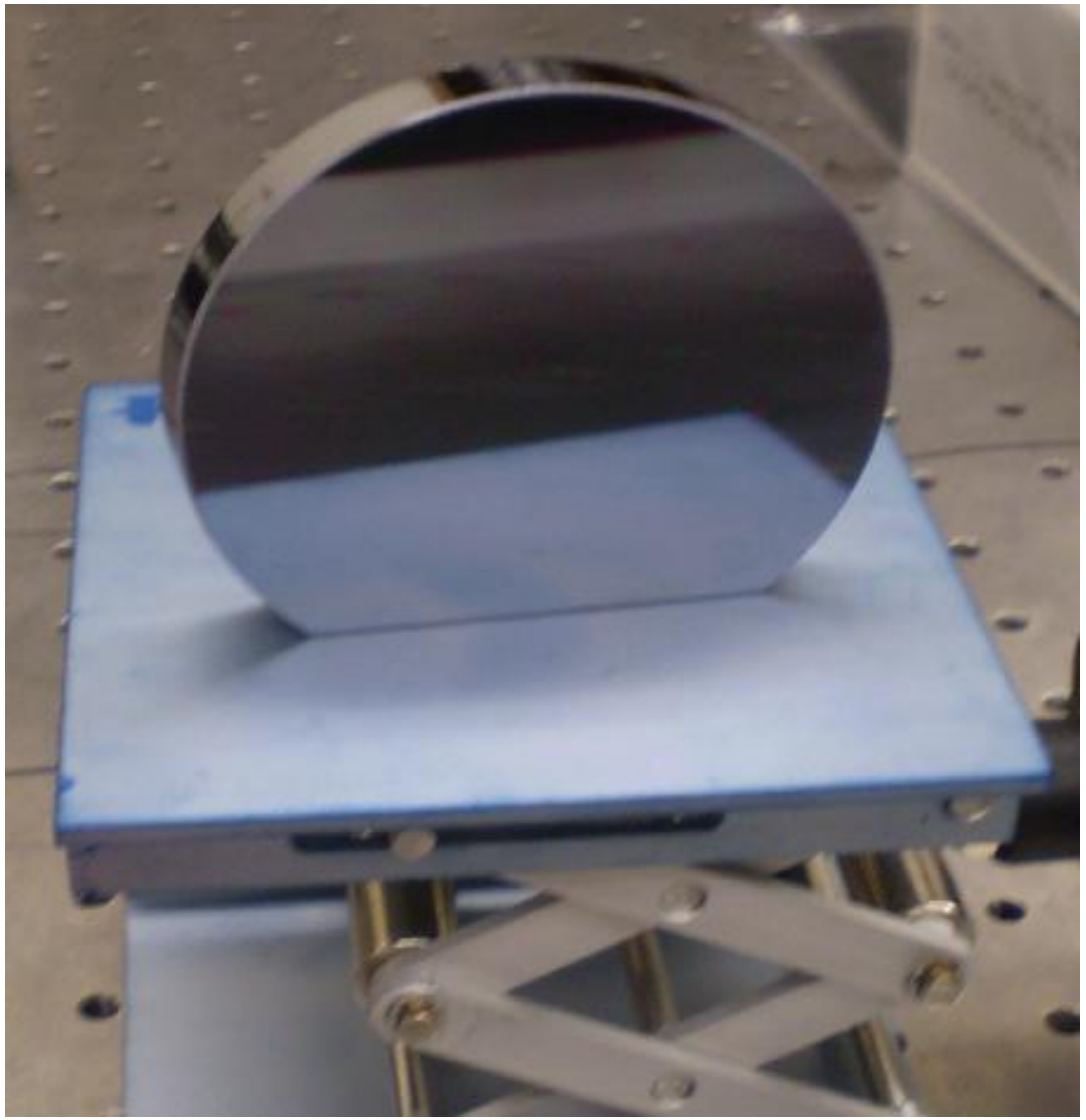
Volume Phase Holographic Gratings diffract light by modulating the index of refraction of a dichromated gelatin. Can be reflective or transmissive. Extra reading: Melso+ 2022

Types of Dispersive Elements: Grisms



From Benjamin Weiner

Grism Example: JWST/NIRCam

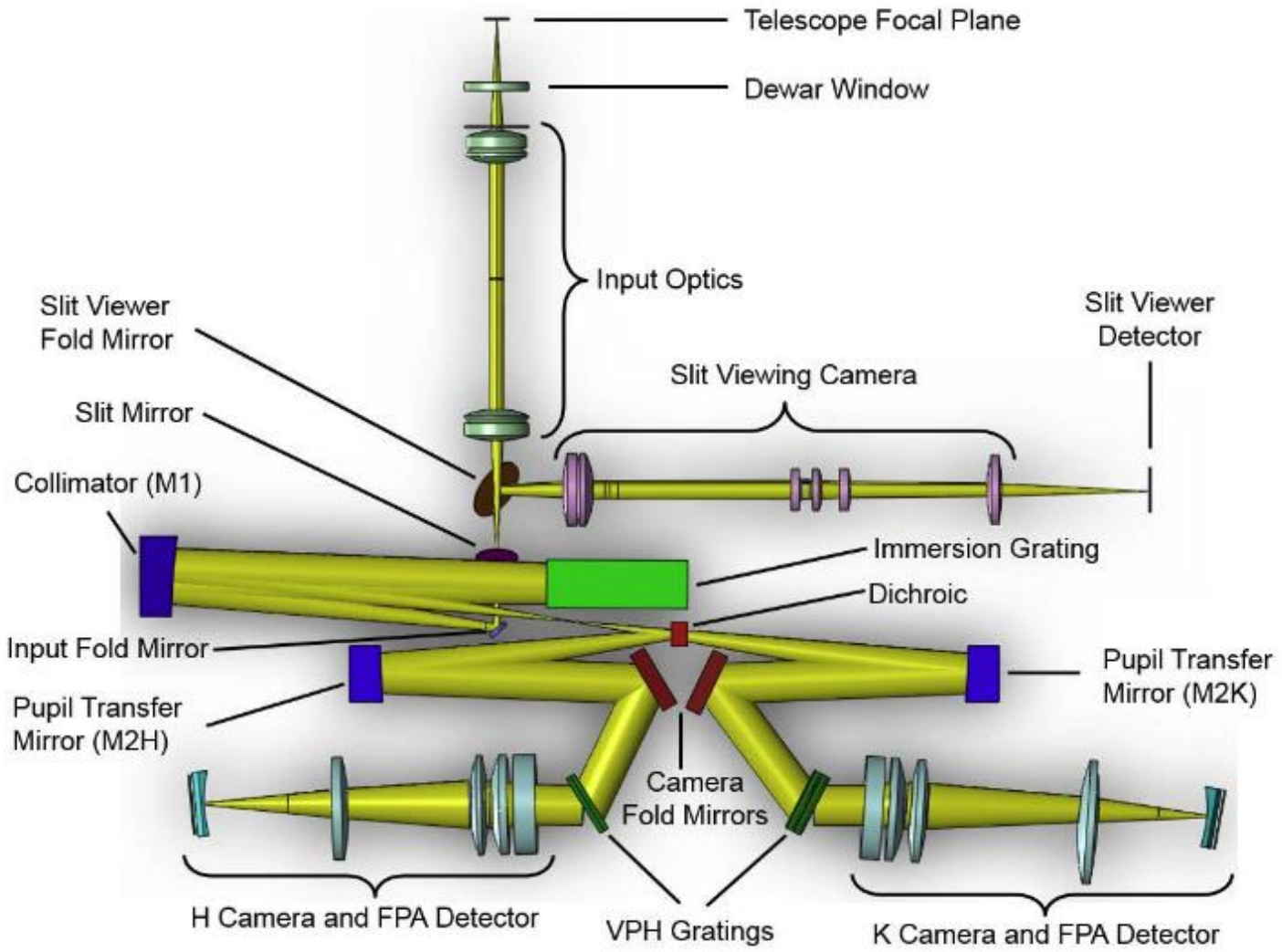


Jaffe+ 2008, Etched grooves into the silicon surface



Jdocs (JWST NIRCam team)

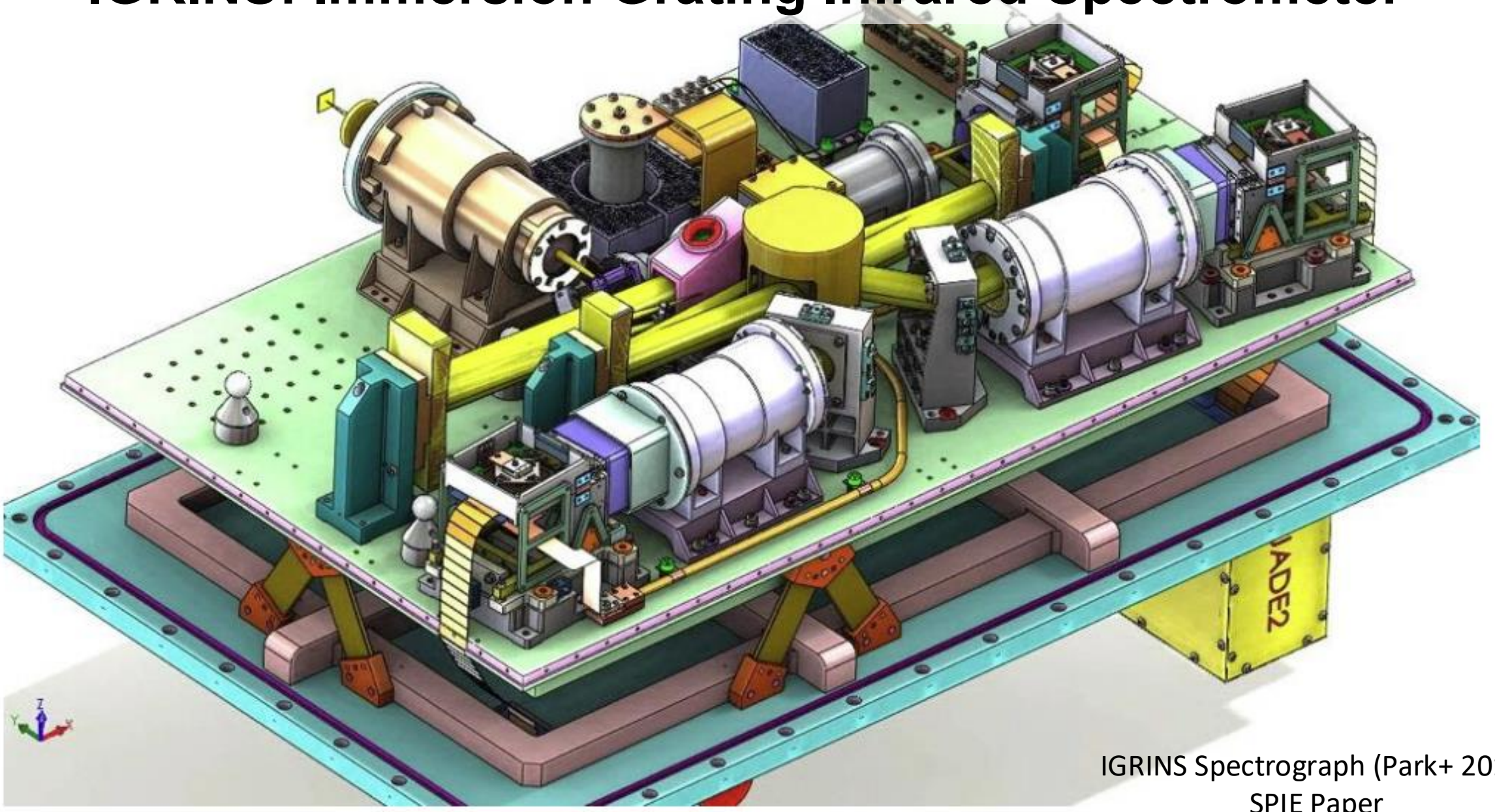
IGRINS: Immersion Grating Infrared Spectrometer



IGRINS uses both Immersion gratings and VPH gratings

IGRINS Spectrograph (Park+ 2014)
SPIE Paper

IGRINS: Immersion Grating Infrared Spectrometer



IGRINS Spectrograph (Park+ 2014)
SPIE Paper